# Analysis in Finite-Dimensional Normed Vector Spaces

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# Coordinate Functions of a Map

## Standing conventions

Throughout, $\mathbb{F}$ denotes either $\mathbb{R}$ or $\mathbb{C}$. A *normed $\mathbb{F}$-vector space* is a pair $(W, \|\cdot\|_W)$ where $W$ is a vector space over $\mathbb{F}$ and $\|\cdot\|_W : W \to [0,\infty)$ satisfies, for all $x, y \in W$ and $\lambda \in \mathbb{F}$,

$$\|x\|_W = 0 \iff x = 0, \qquad \|\lambda x\|_W = |\lambda|\,\|x\|_W, \qquad \|x+y\|_W \le \|x\|_W + \|y\|_W,$$

where $|\cdot|$ is the usual absolute value on $\mathbb{R}$ or the modulus on $\mathbb{C}$.

In this note we assume all vector spaces are *finite-dimensional* over $\mathbb{F}$, so that bases and coordinate maps are available in the usual sense.

## Coordinates in a finite-dimensional space

**Definition 1** (Coordinate functionals associated to a basis)**.** *Let $(W, \|\cdot\|_W)$ be a finite-dimensional normed $\mathbb{F}$-vector space and let*

$$\beta = (w_1, \dots, w_m)$$

*be an ordered basis of $W$. For each $j \in \{1, \dots, m\}$ define the map $\pi_j^\beta : W \to \mathbb{F}$ by the rule: for $y \in W$, write $y$ uniquely as*

$$y = \sum_{k=1}^{m} y_k w_k \qquad (y_k \in \mathbb{F}),$$

*and set*

$$\pi_j^\beta(y) := y_j.$$

*The maps $\pi_1^\beta, \dots, \pi_m^\beta$ are called the* coordinate functionals *of $W$ with respect to $\beta$.*

**Proposition 1** (Basic properties of coordinate functionals)**.** *With notation as above:*

1. *Each $\pi_j^\beta : W \to \mathbb{F}$ is $\mathbb{F}$-linear.*
2. *Every $y \in W$ satisfies the reconstruction formula*

$$y = \sum_{j=1}^{m} \pi_j^\beta(y)\, w_j.$$

*Proof.*
(1) Let $y, z \in W$ and $\lambda \in \mathbb{F}$. Write $y = \sum_{k=1}^m y_k w_k$ and $z = \sum_{k=1}^m z_k w_k$. Then $y+z = \sum_{k=1}^m (y_k + z_k) w_k$ and $\lambda y = \sum_{k=1}^m (\lambda y_k) w_k$ because vector space addition and scalar multiplication distribute over finite sums: one uses

$$(\alpha + \beta) w = \alpha w + \beta w$$

to combine the $k$-th terms in $y + z$, and one uses

$$\lambda(u+v) = \lambda u + \lambda v \quad \text{together with} \quad \lambda(\alpha w) = (\lambda\alpha) w$$

to distribute $\lambda$ over the finite linear combination. By definition of $\pi_j^\beta$ this gives

$$\pi_j^\beta(y+z) = y_j + z_j = \pi_j^\beta(y) + \pi_j^\beta(z), \qquad \pi_j^\beta(\lambda y) = \lambda y_j = \lambda\, \pi_j^\beta(y),$$

so $\pi_j^\beta$ is linear.

(2) If $y = \sum_{k=1}^m y_k w_k$, then $\pi_j^\beta(y) = y_j$ for each $j$, hence $\sum_{j=1}^m \pi_j^\beta(y)\, w_j = \sum_{j=1}^m y_j w_j = y$. $\square$

**Remark 1** (Continuity)**.** *Since $W$ is finite-dimensional, every linear map from $W$ to $\mathbb{F}$ is continuous (equivalently, bounded). In particular, each coordinate functional $\pi_j^\beta$ is continuous.*

## Component functions of a map

**Definition 2** (Component (coordinate) functions of a map)**.** *Let $(V, \|\cdot\|_V)$ and $(W, \|\cdot\|_W)$ be normed $\mathbb{F}$-vector spaces, with $W$ finite-dimensional, and let $f : V \to W$ be a map. Fix an ordered basis $\beta = (w_1, \ldots, w_m)$ of $W$ and let $\pi_1^\beta, \ldots, \pi_m^\beta : W \to \mathbb{F}$ be the associated coordinate functionals. For each $j \in \{1, \ldots, m\}$ define the $j$-th* component function of $f$ with respect to $\beta$ *by*

$$f_j := \pi_j^\beta \circ f : V \to \mathbb{F}.$$

**Proposition 2** (Reconstruction of $f$ from its component functions)**.** *With the notation above, for every $x \in V$,*

$$f(x) = \sum_{j=1}^{m} f_j(x)\, w_j.$$

*Proof.* Fix $x \in V$ and set $y := f(x) \in W$. By the reconstruction formula in $W$,

$$y = \sum_{j=1}^{m} \pi_j^\beta(y)\, w_j.$$

Substituting $y = f(x)$ and using $f_j = \pi_j^\beta \circ f$ gives

$$f(x) = \sum_{j=1}^{m} \pi_j^\beta(f(x))\, w_j = \sum_{j=1}^{m} f_j(x)\, w_j.$$

□

**Remark 2** (Real vs complex scalars)**.** *If $\mathbb{F} = \mathbb{R}$ then each component function satisfies $f_j : V \to \mathbb{R}$. If $\mathbb{F} = \mathbb{C}$ then each component function satisfies $f_j : V \to \mathbb{C}$. In general, the component functions take values in the base field $\mathbb{F}$, not necessarily in $\mathbb{R}$.*

## Example: Reconstruction in $W$ and Reconstruction of $f$

Let $\mathbb{F} = \mathbb{R}$. Let $V = \mathbb{R}^2$ and $W = \mathbb{R}^2$ with the Euclidean norm. Fix the ordered basis

$$\beta = (w_1, w_2) \quad \text{of } W, \qquad w_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad w_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

For $y \in W$, write uniquely

$$y = y_1 w_1 + y_2 w_2,$$

and define the coordinate functionals $\pi_1^\beta, \pi_2^\beta : W \to \mathbb{R}$ by

$$\pi_1^\beta(y) = y_1, \qquad \pi_2^\beta(y) = y_2.$$

**Example 1** (Reconstruction of a vector $y \in W$)**.** *Let*

$$y = \begin{bmatrix} 3 \\ 1 \end{bmatrix} \in W.$$

*We find its coordinates with respect to $\beta$ by solving*

$$\begin{bmatrix} 3 \\ 1 \end{bmatrix} = y_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + y_2 \begin{bmatrix} 1 \\ -1 \end{bmatrix} \implies \begin{bmatrix} 1 & 1 & 3 \\ 1 & 1 & -1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 1 \end{bmatrix}$$

*Thus the system*

$$y_1 + y_2 = 3, \qquad y_1 - y_2 = 1$$

*gives* $y_1 = 2$ *and* $y_2 = 1$. *Hence*

$$\pi_1^\beta(y) = 2, \qquad \pi_2^\beta(y) = 1,$$

*and the reconstruction formula in* $W$ *reads*

$$y = \pi_1^\beta(y)\, w_1 + \pi_2^\beta(y)\, w_2 = 2w_1 + 1w_2 = 2\begin{bmatrix}1\\1\end{bmatrix} + \begin{bmatrix}1\\-1\end{bmatrix} = \begin{bmatrix}3\\1\end{bmatrix}.$$

**Example 2** (Reconstruction of a map $f : V \to W$ from component functions)**.** *Define a map* $f : V \to W$ *by*

$$f(x_1, x_2) = \begin{bmatrix}x_1 + x_2\\x_1 - x_2\end{bmatrix}.$$

*For each* $j \in \{1, 2\}$ *define the component function with respect to* $\beta$

$$f_j := \pi_j^\beta \circ f : V \to \mathbb{R}.$$

*To compute* $f_1, f_2$ *explicitly, fix* $(x_1, x_2) \in V$ *and write* $f(x_1, x_2) = a_1 w_1 + a_2 w_2$. *That is, solve*

$$\begin{bmatrix}x_1 + x_2\\x_1 - x_2\end{bmatrix} = a_1\begin{bmatrix}1\\1\end{bmatrix} + a_2\begin{bmatrix}1\\-1\end{bmatrix} \implies \begin{bmatrix}1 & 1 & x_1 + x_2\\1 & -1 & x_1 - x_2\end{bmatrix} \sim \begin{bmatrix}1 & 0 & x_2\\0 & 1 & x_2\end{bmatrix}$$

*Thus the system*

$$a_1 + a_2 = x_1 + x_2, \qquad a_1 - a_2 = x_1 - x_2,$$

*gives* $a_1 = x_1$ *and* $a_2 = x_2$. *Therefore,*

$$f_1(x_1, x_2) = \pi_1^\beta(f(x_1, x_2)) = a_1 = x_1, \qquad f_2(x_1, x_2) = \pi_2^\beta(f(x_1, x_2)) = a_2 = x_2.$$

*The reconstruction formula for* $f$ *with respect to* $\beta$ *is then, for all* $(x_1, x_2) \in V$,

$$f(x_1, x_2) = f_1(x_1, x_2)\, w_1 + f_2(x_1, x_2)\, w_2 = x_1\begin{bmatrix}1\\1\end{bmatrix} + x_2\begin{bmatrix}1\\-1\end{bmatrix} = \begin{bmatrix}x_1 + x_2\\x_1 - x_2\end{bmatrix}.$$

# The Fréchet Derivative

## Standing conventions

Throughout, $\mathbb{F}$ denotes $\mathbb{R}$. A *normed* $\mathbb{F}$*-vector space* is a pair $(W, \|\cdot\|_W)$ where $W$ is a vector space over $\mathbb{F}$ and $\|\cdot\|_W : W \to [0, \infty)$ satisfies, for all $x, y \in W$ and $\lambda \in \mathbb{F}$,

$$\|x\|_W = 0 \iff x = 0, \qquad \|\lambda x\|_W = |\lambda|\, \|x\|_W, \qquad \|x + y\|_W \le \|x\|_W + \|y\|_W,$$

where $|\cdot|$ is the usual absolute value on $\mathbb{R}$.

All limits are taken with respect to the metrics induced by the norms. In particular, for a map $f : \Omega \to W$ and a point $a \in \Omega$, the expression $x \to a$ means $\|x - a\|_V \to 0$.

In this note, $\Omega \subseteq V$ will always denote an *open* subset of a normed space $(V, \|\cdot\|_V)$.

## Linear approximation and remainder

**Definition 3** (Remainder associated to a candidate linear approximation)**.** *Let $(V, \|\cdot\|_V)$ and $(W, \|\cdot\|_W)$ be normed $\mathbb{R}$-vector spaces, let $\Omega \subseteq V$ be open, let $f : \Omega \to W$, and let $a \in \Omega$. Let $A : V \to W$ be a linear map. Define the* remainder (error) of $f$ at $a$ relative to $A$ *to be the map*

$$r : \{h \in V : a + h \in \Omega\} \to W, \qquad r(h) := f(a + h) - f(a) - A(h).$$

*Equivalently, for $x \in \Omega$ one may write*

$$\mathrm{Err}(x) := f(x) - f(a) - A(x - a),$$

*and then $\mathrm{Err}(a + h) = r(h)$ whenever $a + h \in \Omega$.*

**Remark 3** (The identity $f(a + h) = f(a) + A(h) + r(h)$ is definitional)**.** *For every $h$ with $a + h \in \Omega$, one has*

$$f(a + h) = f(a) + A(h) + r(h).$$

*This is not an additional assumption: it is simply a rewriting of the definition of $r(h)$.*

## Fréchet differentiability

**Definition 4** (Fréchet differentiability at a point)**.** *Let $(V, \|\cdot\|_V)$ and $(W, \|\cdot\|_W)$ be normed $\mathbb{R}$-vector spaces, let $\Omega \subseteq V$ be open, let $f : \Omega \to W$, and let $a \in \Omega$. We say that $f$ is* Fréchet differentiable at $a$ *if there exists a bounded linear map $A : V \to W$ such that*

$$\lim_{h \to 0} \frac{\|r(h)\|_W}{\|h\|_V} = 0,$$

*that is,*

$$\lim_{h \to 0} \frac{\|f(a + h) - f(a) - A(h)\|_W}{\|h\|_V} = 0.$$

*In that case, the map $A$ is unique and is denoted by $(Df)_a$; it is called the* Fréchet derivative of $f$ at $a$.

**Remark 4** ($\varepsilon$–$\delta$ form of the definition)**.** *Let $f : \Omega \to W$ and $a \in \Omega$. The condition*

$$\lim_{h \to 0} \frac{\|f(a + h) - f(a) - A(h)\|_W}{\|h\|_V} = 0$$

*is equivalent to the following statement: for every $\varepsilon > 0$ there exists $\delta > 0$ such that for all $h \in V$,*

$$0 < \|h\|_V < \delta \quad \text{and} \quad a + h \in \Omega \quad \Longrightarrow \quad \|f(a + h) - f(a) - A(h)\|_W \le \varepsilon \, \|h\|_V.$$

*In particular, the definition asserts that the remainder is bounded by an arbitrarily small multiple of $\|h\|_V$ for all sufficiently small $h$.*

## Linear maps are Fréchet differentiable

**Proposition 3** (Linear maps are Fréchet differentiable)**.** *Let $(V, \|\cdot\|_V)$ and $(W, \|\cdot\|_W)$ be normed $\mathbb{R}$-vector spaces, and let $L \in B(V, W)$. Then $L$ is Fréchet differentiable at every $a \in V$, and*

$$(DL)_a = L \qquad (a \in V).$$

*Proof.*
Fix $a \in V$. Consider the bounded linear map $A : V \to W$ defined by $A := L$. For $h \in V$,

$$L(a+h) - L(a) - A(h) = L(a+h) - L(a) - L(h) = 0,$$

using linearity of $L$. Therefore, for every $h \neq 0$,

$$\frac{\|L(a+h) - L(a) - A(h)\|_W}{\|h\|_V} = \frac{\|0\|_W}{\|h\|_V} = 0,$$

so the limit as $h \to 0$ is 0. Hence $L$ is Fréchet differentiable at $a$ with $(DL)_a = A = L$. □

**Proposition 4** (Why the denominator matters)**.** *Let $(V, \|\cdot\|_V)$ and $(W, \|\cdot\|_W)$ be normed $\mathbb{R}$-vector spaces, let $\Omega \subseteq V$ be open, let $f : \Omega \to W$, and let $a \in \Omega$. Fix a linear map $A : V \to W$ and define $r(h) = f(a+h) - f(a) - A(h)$. The implication*

$$\big( \ \|r(h)\|_W \to 0 \ \textit{as} \ h \to 0 \ \big) \implies \left( \frac{\|r(h)\|_W}{\|h\|_V} \to 0 \ \textit{as} \ h \to 0 \right)$$

*is false in general.*

*Proof.*
Take $V = W = \mathbb{R}$ with the usual absolute value as the norm, and define a function $r : \mathbb{R} \to \mathbb{R}$ by $r(h) = |h|$. Then $r(h) \to 0$ as $h \to 0$, but for every $h \neq 0$,

$$\frac{|r(h)|}{|h|} = \frac{|h|}{|h|} = 1,$$

so the quotient does not tend to 0. □

**Remark 5** (Conceptual summary)**.** *Let $f : \Omega \to W$ and $a \in \Omega$. Fréchet differentiability at $a$ asserts the existence of a bounded linear map $(Df)_a : V \to W$ such that, for small increments $h$,*

$$f(a+h) = f(a) + (Df)_a(h) + r(h), \qquad r(h) := f(a+h) - f(a) - (Df)_a(h),$$

*with the quantitative control*

$$\frac{\|r(h)\|_W}{\|h\|_V} \to 0 \qquad (h \to 0).$$

*Thus $f(a) + (Df)_a(h)$ provides the correct* first-order (linear) approximation *of $f$ near $a$, and the remaining term is negligible compared with $\|h\|_V$.*

## Example: A Simple Computation in $\mathbb{R}^2$

**Example 3** (A function with derivative 0 at the origin)**.** *Let $V = \mathbb{R}^2$ and $W = \mathbb{R}$ with the Euclidean norm on $\mathbb{R}^2$ and the absolute value on $\mathbb{R}$. Define $f : \mathbb{R}^2 \to \mathbb{R}$ by*

$$f(x_1, x_2) = x_1^2 + x_2^2,$$

*and let $a = (0, 0)$. We claim that $f$ is Fréchet differentiable at $a$ and that $(Df)_a$ is the zero linear map.*

*Let $A : \mathbb{R}^2 \to \mathbb{R}$ be the zero map, so $A(h) = 0$ for all $h \in \mathbb{R}^2$. For $h = (h_1, h_2)$ we have $a + h = h$ and $f(a) = 0$, so*

$$f(a + h) - f(a) - A(h) = f(h_1, h_2) = h_1^2 + h_2^2.$$

*Hence, for $h \neq 0$,*

$$\frac{|f(a + h) - f(a) - A(h)|}{\|h\|_2} = \frac{h_1^2 + h_2^2}{\sqrt{h_1^2 + h_2^2}} = \sqrt{h_1^2 + h_2^2} = \|h\|_2 \longrightarrow 0 \quad (h \to 0).$$

*Therefore $f$ is Fréchet differentiable at $a$ and $(Df)_a = A$, i.e. $(Df)_{(0,0)}$ is the zero linear map.*

# The One-Variable Derivative

## Standing conventions

Throughout, $\mathbb{F}$ denotes $\mathbb{R}$. A *normed $\mathbb{F}$-vector space* is a pair $(W, \|\cdot\|_W)$ where $W$ is a vector space over $\mathbb{F}$ and $\|\cdot\|_W : W \to [0, \infty)$ satisfies, for all $x, y \in W$ and $\lambda \in \mathbb{F}$,

$$\|x\|_W = 0 \iff x = 0, \qquad \|\lambda x\|_W = |\lambda|\, \|x\|_W, \qquad \|x + y\|_W \leq \|x\|_W + \|y\|_W,$$

where $|\cdot|$ is the usual absolute value on $\mathbb{R}$.

All limits in $W$ are taken with respect to the norm $\|\cdot\|_W$. In particular, for a function $q : J \to W$ and a point $t_0 \in J$, the statement

$$\lim_{t \to t_0} q(t) = w$$

means

$$\lim_{t \to t_0} \|q(t) - w\|_W = 0.$$

## The one-variable derivative

**Definition 5** (Derivative at a point)**.** *Let $I \subseteq \mathbb{R}$ be an open interval, let $(W, \|\cdot\|_W)$ be a normed $\mathbb{R}$-vector space, and let $g : I \to W$ be a map. Fix $t_0 \in I$. If there exists $w \in W$ such that*

$$\lim_{h \to 0} \left\| \frac{g(t_0 + h) - g(t_0)}{h} - w \right\|_W = 0,$$

*then we say that $g$ is* differentiable at $t_0$ *and we define the* derivative of $g$ at $t_0$ *by*

$$g'(t_0) := w.$$

**Remark 6** (Equivalent limit notation)**.** *The condition*

$$\lim_{h\to 0}\left\|\frac{g(t_0+h)-g(t_0)}{h}-w\right\|_W=0$$

*is equivalent to the statement*

$$\lim_{h\to 0}\frac{g(t_0+h)-g(t_0)}{h}=w,$$

*where the latter limit is understood as a limit in the normed space $W$.*

**Proposition 5** (First-order approximation in one variable)**.** *Let $I\subseteq\mathbb{R}$ be open, let $g:I\to W$, and let $t_0\in I$. Then $g$ is differentiable at $t_0$ with $g'(t_0)=w$ if and only if there exists a map*

$$r:\{h\in\mathbb{R}:t_0+h\in I\}\to W$$

*such that for all $h$ with $t_0+h\in I$,*

$$g(t_0+h)=g(t_0)+h\,w+r(h),$$

*and*

$$\lim_{h\to 0}\frac{\|r(h)\|_W}{|h|}=0.$$

*Proof.*
Assume $g$ is differentiable at $t_0$ with derivative $w$. Define $r(h):=g(t_0+h)-g(t_0)-h\,w$. Then the identity

$$g(t_0+h)=g(t_0)+h\,w+r(h)$$

holds by definition of $r(h)$. For $h\neq 0$,

$$\frac{\|r(h)\|_W}{|h|}=\left\|\frac{g(t_0+h)-g(t_0)-h\,w}{h}\right\|_W=\left\|\frac{g(t_0+h)-g(t_0)}{h}-w\right\|_W,$$

which tends to 0 by the definition of differentiability. Hence $\|r(h)\|_W/|h|\to 0$.

Conversely, assume there is a map $r$ such that $g(t_0+h)=g(t_0)+h\,w+r(h)$ and $\|r(h)\|_W/|h|\to 0$. Then for $h\neq 0$,

$$\frac{g(t_0+h)-g(t_0)}{h}-w=\frac{r(h)}{h},$$

so

$$\left\|\frac{g(t_0+h)-g(t_0)}{h}-w\right\|_W=\frac{\|r(h)\|_W}{|h|}\longrightarrow 0.$$

Thus $g$ is differentiable at $t_0$ and $g'(t_0)=w$. □

## Fréchet differentiability on $\mathbb{R}$

**Definition 6** (Fréchet differentiability for maps $I\subseteq\mathbb{R}\to W$)**.** *Let $I\subseteq\mathbb{R}$ be open and let $g:I\to W$. Fix $t_0\in I$. We say that $g$ is* Fréchet differentiable at $t_0$ *if there exists a bounded linear map $A:\mathbb{R}\to W$ such that*

$$\lim_{h\to 0}\frac{\|g(t_0+h)-g(t_0)-A(h)\|_W}{|h|}=0.$$

*In that case $A$ is unique and is denoted by $(Dg)_{t_0}$.*

**Proposition 6** (Linear maps $\mathbb{R} \to W$)**.** *Let $A : \mathbb{R} \to W$ be linear. Then for all $h \in \mathbb{R}$,*

$$A(h) = h\,A(1).$$

*In particular, every linear map $A : \mathbb{R} \to W$ is determined by the single vector $A(1) \in W$.*

*Proof.*
For $h \in \mathbb{R}$ one has $h = h \cdot 1$, hence by linearity,

$$A(h) = A(h \cdot 1) = h\,A(1).$$

□

## Connection to Fréchet differentiability

**Proposition 7** (Equivalence in one variable)**.** *Let $I \subseteq \mathbb{R}$ be open, let $g : I \to W$, and let $t_0 \in I$. Then the following are equivalent:*

1. *$g$ is differentiable at $t_0$ in the one-variable sense, and $g'(t_0)$ exists in $W$.*
2. *$g$ is Fréchet differentiable at $t_0$ as a map $I \subseteq \mathbb{R} \to W$.*

*Moreover, if these conditions hold, then the Fréchet derivative $(Dg)_{t_0} : \mathbb{R} \to W$ is the linear map*

$$(Dg)_{t_0}(h) = h\,g'(t_0) \qquad (h \in \mathbb{R}),$$

*and conversely,*

$$g'(t_0) = (Dg)_{t_0}(1).$$

*Proof.*
(1)⇒(2). Assume $g$ is differentiable at $t_0$ and set $w := g'(t_0)$. Define a linear map $A : \mathbb{R} \to W$ by $A(h) := h\,w$. By Proposition 4, the differentiability of $g$ at $t_0$ is equivalent to the existence of a remainder $r(h)$ such that

$$g(t_0 + h) = g(t_0) + h\,w + r(h) \quad \text{and} \quad \frac{\|r(h)\|_W}{|h|} \to 0.$$

But then $g(t_0 + h) - g(t_0) - A(h) = r(h)$, hence

$$\frac{\|g(t_0 + h) - g(t_0) - A(h)\|_W}{|h|} = \frac{\|r(h)\|_W}{|h|} \to 0.$$

Thus $g$ is Fréchet differentiable at $t_0$ with $(Dg)_{t_0} = A$, i.e. $(Dg)_{t_0}(h) = h\,g'(t_0)$.

(2)⇒(1). Assume $g$ is Fréchet differentiable at $t_0$. Then there exists a linear map $A : \mathbb{R} \to W$ such that

$$\frac{\|g(t_0 + h) - g(t_0) - A(h)\|_W}{|h|} \to 0.$$

By Proposition 5, $A(h) = h\,A(1)$. Set $w := A(1)$, so $A(h) = h\,w$. Then

$$\frac{\|g(t_0 + h) - g(t_0) - h\,w\|_W}{|h|} \to 0,$$

which is equivalent to

$$\left\| \frac{g(t_0 + h) - g(t_0)}{h} - w \right\|_W \to 0.$$

Hence $g$ is differentiable at $t_0$ and $g'(t_0) = w = A(1) = (Dg)_{t_0}(1)$. □

**Remark 7** (Interpretation)**.** *In one variable, the usual derivative $g'(t_0) \in W$ and the Fréchet derivative $(Dg)_{t_0} : \mathbb{R} \to W$ carry the same information. The scalar increment $h \in \mathbb{R}$ is converted into an output increment in $W$ by multiplication with the vector $g'(t_0)$, namely*

$$(Dg)_{t_0}(h) = h\, g'(t_0).$$

## Example: A Computation in $W = \mathbb{R}^2$

**Example 4** (A linear example)**.** *Let $W = \mathbb{R}^2$ with the Euclidean norm and let $I = \mathbb{R}$. Define $g : \mathbb{R} \to \mathbb{R}^2$ by*

$$g(t) = \begin{bmatrix} t \\ 2t \end{bmatrix}.$$

*Fix $t_0 \in \mathbb{R}$. For $h \neq 0$,*

$$\frac{g(t_0 + h) - g(t_0)}{h} = \frac{1}{h}\left(\begin{bmatrix} t_0 + h \\ 2(t_0 + h) \end{bmatrix} - \begin{bmatrix} t_0 \\ 2t_0 \end{bmatrix}\right) = \begin{bmatrix} 1 \\ 2 \end{bmatrix}.$$

*Thus the limit exists and*

$$g'(t_0) = \begin{bmatrix} 1 \\ 2 \end{bmatrix}.$$

*By Proposition 6, the Fréchet derivative $(Dg)_{t_0} : \mathbb{R} \to \mathbb{R}^2$ is given by*

$$(Dg)_{t_0}(h) = h\, g'(t_0) = h \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} h \\ 2h \end{bmatrix}.$$

## Directional Derivatives

### Standing conventions

Throughout, $\mathbb{F}$ denotes $\mathbb{R}$. A *normed $\mathbb{F}$-vector space* is a pair $(W, \|\cdot\|_W)$ where $W$ is a vector space over $\mathbb{F}$ and $\|\cdot\|_W : W \to [0, \infty)$ satisfies, for all $x, y \in W$ and $\lambda \in \mathbb{F}$,

$$\|x\|_W = 0 \iff x = 0, \qquad \|\lambda x\|_W = |\lambda|\, \|x\|_W, \qquad \|x + y\|_W \leq \|x\|_W + \|y\|_W,$$

where $|\cdot|$ is the usual absolute value on $\mathbb{R}$.

Let $(V, \|\cdot\|_V)$ and $(W, \|\cdot\|_W)$ be normed $\mathbb{R}$-vector spaces. Let $\Omega \subseteq V$ be open, and let $f : \Omega \to W$ be a map. Fix $a \in \Omega$.

All limits in $W$ are taken with respect to $\|\cdot\|_W$. In particular, for a function $q : J \to W$ and a point $t_0 \in J$, the statement

$$\lim_{t \to t_0} q(t) = w$$

means

$$\lim_{t \to t_0} \|q(t) - w\|_W = 0.$$

## Restricting $f$ to a line through $a$

**Proposition 8** (Local existence of a line segment in $\Omega$)**.** *Let $v \in V$. Since $\Omega$ is open and $a \in \Omega$, there exists $r > 0$ such that $B(a, r) \subseteq \Omega$. If $v \neq 0$, set*

$$I := \left(-\frac{r}{\|v\|_V}, \frac{r}{\|v\|_V}\right).$$

*If $v = 0$, set $I := (-1, 1)$. Then $I$ is an open interval containing $0$ and*

$$a + tv \in \Omega \qquad \text{for all } t \in I.$$

*Proof.*
Choose $r > 0$ such that $B(a, r) \subseteq \Omega$.

If $v \neq 0$ and $t \in I$, then $|t|\,\|v\|_V < r$. Hence

$$\|a + tv - a\|_V = \|tv\|_V = |t|\,\|v\|_V < r,$$

so $a + tv \in B(a, r) \subseteq \Omega$.

If $v = 0$ then $a + tv = a \in \Omega$ for all $t$, so any open interval containing $0$ works; in particular $(-1, 1)$ does. □

**Definition 7** (Restriction of $f$ to the line through $a$ in direction $v$)**.** *Let $v \in V$ and let $I$ be an open interval containing $0$ such that $a + tv \in \Omega$ for all $t \in I$ (for instance, $I$ as in Proposition 8). Define*

$$g : I \to W, \qquad g(t) := f(a + tv).$$

## Directional derivatives

**Definition 8** (Directional derivative)**.** *Let $v \in V$ and let $g : I \to W$ be defined by $g(t) = f(a+tv)$ on an interval $I$ as above. If $g$ is differentiable at $0$, we define the* directional derivative of $f$ at $a$ in direction $v$ *to be*

$$(D_v f)_a := g'(0).$$

*Equivalently, if the limit exists in $W$, one has*

$$(D_v f)_a = \lim_{t \to 0} \frac{f(a + tv) - f(a)}{t}.$$

**Remark 8** (First-order approximation along the line)**.** *If $(D_v f)_a$ exists, then there is a remainder $\rho(t) \in W$ such that for all $t$ sufficiently close to $0$,*

$$f(a + tv) = f(a) + t\,(D_v f)_a + \rho(t),$$

*and*

$$\lim_{t \to 0} \frac{\|\rho(t)\|_W}{|t|} = 0.$$

*Thus the directional derivative captures the first-order behaviour of $f$ along the straight line $t \mapsto a + tv$.*

**Proposition 9** (Two basic properties)**.** *Assume $(D_v f)_a$ exists.*

1. *If $v = 0$, then $(D_0 f)_a = 0$.*

*2. For every $\lambda \in \mathbb{R}$, one has $(D_{\lambda v} f)_a = \lambda\,(D_v f)_a$.*

*Proof.*
(1) If $v = 0$ then $f(a+tv) = f(a)$ for all $t$, so the difference quotient is 0 for all $t \neq 0$ and hence the limit is 0.

(2) Let $\lambda \in \mathbb{R}$. If $\lambda = 0$ this is part (1). Assume $\lambda \neq 0$. For $t \neq 0$ set $s = \lambda t$, so that $s \to 0$ as $t \to 0$. Then

$$\frac{f(a + t(\lambda v)) - f(a)}{t} = \frac{f(a + sv) - f(a)}{s/\lambda} = \lambda \frac{f(a + sv) - f(a)}{s}.$$

Taking limits as $t \to 0$ (equivalently $s \to 0$) gives $(D_{\lambda v} f)_a = \lambda (D_v f)_a$. □

## Connection to Fréchet differentiability

**Proposition 10** (Fréchet differentiability implies existence of all directional derivatives)**.** *Assume $f$ is Fréchet differentiable at $a$, with Fréchet derivative $(Df)_a : V \to W$. Then for every $v \in V$, the directional derivative $(D_v f)_a$ exists and satisfies*

$$(D_v f)_a = (Df)_a(v).$$

*Proof.*
Since $f$ is Fréchet differentiable at $a$, there exists a remainder $r : \{h \in V : a + h \in \Omega\} \to W$ such that

$$f(a + h) = f(a) + (Df)_a(h) + r(h), \qquad \lim_{h \to 0} \frac{\|r(h)\|_W}{\|h\|_V} = 0.$$

Fix $v \in V$. For $t$ sufficiently close to 0 we have $a + tv \in \Omega$ and thus

$$f(a + tv) - f(a) = (Df)_a(tv) + r(tv) = t\,(Df)_a(v) + r(tv).$$

For $t \neq 0$,

$$\frac{f(a + tv) - f(a)}{t} = (Df)_a(v) + \frac{r(tv)}{t}.$$

Moreover,

$$\left\| \frac{r(tv)}{t} \right\|_W = \frac{\|r(tv)\|_W}{|t|} = \frac{\|r(tv)\|_W}{\|tv\|_V}\,\|v\|_V \longrightarrow 0 \cdot \|v\|_V = 0,$$

because $\|tv\|_V = |t|\,\|v\|_V$ and $\|r(h)\|_W / \|h\|_V \to 0$ as $h \to 0$. Hence the difference quotient converges to $(Df)_a(v)$, so $(D_v f)_a$ exists and equals $(Df)_a(v)$. □

**Remark 9** (Why the converse can fail)**.** *Directional derivatives are defined by restricting $f$ to one line at a time. Consequently, even if $(D_v f)_a$ exists for every $v \in V$, $f$ may fail to be Fréchet differentiable at $a$. Fréchet differentiability requires a single bounded linear map $A : V \to W$ such that*

$$\frac{\|f(a + h) - f(a) - A(h)\|_W}{\|h\|_V} \to 0 \qquad (h \to 0),$$

*which is a uniform first-order approximation (valid for all small increments $h$ simultaneously). Equivalently, the condition above is the $\varepsilon$–$\delta$ statement: for every $\varepsilon > 0$ there exists $\delta > 0$ such that for all $h \in V$,*

$$0 < \|h\|_V < \delta \;\text{ and }\; a + h \in \Omega \quad \Longrightarrow \quad \|f(a + h) - f(a) - A(h)\|_W \leq \varepsilon\,\|h\|_V.$$

*In this sense, once $\varepsilon$ is fixed, the same $\delta$ works simultaneously for every direction and every vector $h$ with $\|h\|_V < \delta$; there is no choice of $\delta$ that depends on the direction. Directional differentiability provides only line-by-line information and need not assemble into such a single linear approximation.*

## Example: Directional derivatives exist but no Fréchet derivative

**Example 5** (A standard counterexample in $\mathbb{R}^2$)**.** *Let $V = \mathbb{R}^2$ with the Euclidean norm and let $W = \mathbb{R}$ with the absolute value. Define $f : \mathbb{R}^2 \to \mathbb{R}$ by*

$$f(x_1, x_2) = \begin{cases} \dfrac{x_1^2 x_2}{x_1^4 + x_2^2}, & (x_1, x_2) \neq (0,0), \\ 0, & (x_1, x_2) = (0,0). \end{cases}$$

*Let $a = (0,0)$.*
Directional derivatives at $a$. *Fix $v = (v_1, v_2) \in \mathbb{R}^2$. For $t \neq 0$,*

$$\frac{f(tv_1, tv_2) - f(0,0)}{t} = \frac{1}{t} \cdot \frac{t^3 v_1^2 v_2}{t^4 v_1^4 + t^2 v_2^2} = \frac{v_1^2 v_2}{t^2 v_1^4 + v_2^2}.$$

*If $v_2 \neq 0$ then $t^2 v_1^4 + v_2^2 \to v_2^2$ as $t \to 0$, hence*

$$\lim_{t \to 0} \frac{f(tv_1, tv_2) - f(0,0)}{t} = \frac{v_1^2 v_2}{v_2^2} = \frac{v_1^2}{v_2}.$$

*If $v_2 = 0$ then $f(tv_1, 0) = 0$ for all $t$, hence the difference quotient is $0$ and the limit is $0$. Thus $(D_v f)_{(0,0)}$ exists for every $v \in \mathbb{R}^2$.*

Failure of Fréchet differentiability at $a$. *Assume for contradiction that $f$ is Fréchet differentiable at $a$ with derivative $A : \mathbb{R}^2 \to \mathbb{R}$. By Proposition 9, for every $v$ one has $(D_v f)_a = A(v)$, so $v \mapsto (D_v f)_a$ must be linear. However, using the values computed above,*

$$(D_{(1,1)} f)_{(0,0)} = 1, \qquad (D_{(1,0)} f)_{(0,0)} = 0 \qquad (D_{(2,1)} f)_{(0,0)} = 4$$

*Linearity would therefore imply*

$$(D_{(2,1)} f)_{(0,0)} = A(2,1) = A(1,1) + A(1,0) = (D_{(1,1)} f)_{(0,0)} + (D_{(1,0)} f)_{(0,0)} = 1 + 0 = 1,$$

*contradicting $(D_{(2,1)} f)_{(0,0)} = 4$. Hence no such linear map $A$ exists and $f$ is not Fréchet differentiable at $a$.*

## Partial Derivatives

### Standing conventions

Throughout, $\mathbb{F}$ denotes $\mathbb{R}$. Let $\Omega \subseteq \mathbb{R}^d$ be open and let $(W, \|\cdot\|_W)$ be a normed $\mathbb{R}$-vector space. Let $f : \Omega \to W$ be a map and fix $a = (a_1, \ldots, a_d) \in \Omega$.

For $i \in \{1, \ldots, d\}$, let $e_i \in \mathbb{R}^d$ denote the $i$-th standard basis vector:

$$e_i = (0, \ldots, 0, \underbrace{1}_{i\text{-th position}}, 0, \ldots, 0).$$

All limits in $W$ are taken with respect to $\|\cdot\|_W$.

**Remark 10** (Why partial derivatives are stated on $\mathbb{R}^d$)**.** *Directional derivatives are intrinsic: for a map $f : \Omega \subseteq V \to W$ and a direction $v \in V$, the expression $t \mapsto a + tv$ makes sense in any normed vector space $V$, so $(D_v f)_a$ is defined without choosing coordinates. By contrast, partial derivatives are coordinate notions: the definition of $\partial_i f(a)$ uses the distinguished coordinate directions $e_1, \ldots, e_d$, which are canonically available only when the domain is identified with $\mathbb{R}^d$ (with its standard basis). If $V$ is a general finite-dimensional normed space and one fixes an ordered basis $\beta = (v_1, \ldots, v_d)$ of $V$, then one may define a basis-dependent analogue by*

$$\partial_{v_i} f(a) := (D_{v_i} f)_a \qquad (i = 1, \ldots, d),$$

*but this depends on the chosen basis and is not canonical.*

## Restricting $f$ to a coordinate line

**Proposition 11** (Local existence of coordinate line segments in $\Omega$)**.** *Let $i \in \{1, \ldots, d\}$. Since $\Omega$ is open and $a \in \Omega$, there exists $r > 0$ such that*

$$B(a, r) \subseteq \Omega,$$

*where $B(a, r)$ denotes the open ball in $\mathbb{R}^d$ (with its Euclidean norm). Set*

$$I := (-r, r).$$

*Then $I$ is an interval containing $0$ and*

$$a + te_i \in \Omega \qquad \text{for all } t \in I.$$

*Proof.*
Let $r > 0$ satisfy $B(a, r) \subseteq \Omega$. If $t \in (-r, r)$, then

$$\|a + te_i - a\|_2 = \|te_i\|_2 = |t| < r,$$

so $a + te_i \in B(a, r) \subseteq \Omega$. □

## Partial derivatives

**Definition 9** ($i$-th partial derivative)**.** *Let $i \in \{1, \ldots, d\}$. Let $I$ be an open interval containing $0$ such that $a + te_i \in \Omega$ for all $t \in I$ (for instance, $I$ as in Proposition 10). Define the one-variable map*

$$g_i : I \to W, \qquad g_i(t) := f(a + te_i).$$

*If $g_i$ is differentiable at $0$, we define the $i$-th partial derivative of $f$ at $a$ by*

$$\partial_i f(a) := g_i'(0).$$

*Equivalently, if the limit exists in $W$, one has*

$$\partial_i f(a) = \lim_{t \to 0} \frac{f(a + te_i) - f(a)}{t}.$$

**Remark 11** (Interpretation)**.** *The partial derivative $\partial_i f(a)$ measures the first-order change of $f$ at $a$ when only the $i$-th coordinate is varied and all other coordinates are kept fixed. In particular, it is the derivative of the restriction of $f$ to the coordinate line*

$$t \longmapsto a + te_i.$$

## Connection to directional derivatives

**Proposition 12** (Partial derivatives as special directional derivatives)**.** *Let $i \in \{1, \dots, d\}$. If the directional derivative $(D_{e_i} f)_a$ exists, then the $i$-th partial derivative exists and*

$$\partial_i f(a) = (D_{e_i} f)_a.$$

*Conversely, if the $i$-th partial derivative $\partial_i f(a)$ exists, then the directional derivative in direction $e_i$ exists and equals it.*

*Proof.*

Both quantities are defined by the same one-variable limit:

$$\lim_{t \to 0} \frac{f(a + te_i) - f(a)}{t},$$

provided the limit exists in $W$. Hence they exist simultaneously and are equal. □

## Connection to Fréchet differentiability

**Proposition 13** (Fréchet differentiability implies existence of all partial derivatives)**.** *Assume $f$ is Fréchet differentiable at $a$, with Fréchet derivative $(Df)_a : \mathbb{R}^d \to W$. Then for each $i \in \{1, \dots, d\}$ the partial derivative $\partial_i f(a)$ exists and satisfies*

$$\partial_i f(a) = (Df)_a(e_i).$$

*Proof.*

If $f$ is Fréchet differentiable at $a$, then by the result proved in the section on directional derivatives, all directional derivatives exist and satisfy

$$(D_v f)_a = (Df)_a(v) \qquad (v \in \mathbb{R}^d).$$

Applying this with $v = e_i$ and using Proposition 11 yields

$$\partial_i f(a) = (D_{e_i} f)_a = (Df)_a(e_i).$$

□

**Remark 12** (Partial derivatives are weaker hypiotheses than Fréchet differentiability)**.** *The existence of the partial derivatives $\partial_i f(a)$ for all $i$ is a coordinate-axis condition. It does not, by itself, guarantee that $f$ admits a single first-order linear approximation valid for all small increments $h \in \mathbb{R}^d$, and hence it does not, by itself, imply Fréchet differentiability at $a$.*

# Differentiation Rules

## Standing conventions

Throughout, $\mathbb{F}$ denotes $\mathbb{R}$. Let $(V, \|\cdot\|_V)$ and $(W, \|\cdot\|_W)$ be normed $\mathbb{R}$-vector spaces, and let $\Omega \subseteq V$ be open. When $f : \Omega \to W$ is Fréchet differentiable at $a \in \Omega$, its derivative is denoted by $(Df)_a : V \to W$. All derivatives below are Fréchet derivatives.

## Sum rule and scalar multiple rule

**Proposition 14** (Sum rule)**.** *Let $f, g : \Omega \to W$ and let $a \in \Omega$. If $f$ and $g$ are Fréchet differentiable at $a$, then $f + g$ is Fréchet differentiable at $a$ and*

$$(D(f+g))_a = (Df)_a + (Dg)_a,$$

*where the sum on the right-hand side is the pointwise sum of linear maps $V \to W$.*

*Proof.*
Let $f, g : \Omega \to W$ be Fréchet differentiable at $a$. By definition, there exist bounded linear maps $A : V \to W$ and $B : V \to W$ and maps

$$r_f : \{h \in V : a + h \in \Omega\} \to W, \qquad r_g : \{h \in V : a + h \in \Omega\} \to W$$

such that

$$f(a+h) = f(a) + A(h) + r_f(h), \qquad \lim_{h \to 0} \frac{\|r_f(h)\|_W}{\|h\|_V} = 0,$$

$$g(a+h) = g(a) + B(h) + r_g(h), \qquad \lim_{h \to 0} \frac{\|r_g(h)\|_W}{\|h\|_V} = 0.$$

Then for all $h$ with $a + h \in \Omega$,

$$(f+g)(a+h) = (f+g)(a) + (A+B)(h) + \big(r_f(h) + r_g(h)\big).$$

Moreover,

$$\frac{\|r_f(h) + r_g(h)\|_W}{\|h\|_V} \leq \frac{\|r_f(h)\|_W}{\|h\|_V} + \frac{\|r_g(h)\|_W}{\|h\|_V} \longrightarrow 0.$$

Hence $f + g$ is Fréchet differentiable at $a$ and $(D(f+g))_a = A + B = (Df)_a + (Dg)_a$. □

**Proposition 15** (Scalar multiple rule)**.** *Let $f : \Omega \to W$, let $a \in \Omega$, and let $\lambda \in \mathbb{R}$. If $f$ is Fréchet differentiable at $a$, then $\lambda f$ is Fréchet differentiable at $a$ and*

$$(D(\lambda f))_a = \lambda (Df)_a.$$

*Proof.*
Let $f : \Omega \to W$ be Fréchet differentiable at $a$, and write

$$f(a+h) = f(a) + A(h) + r(h), \qquad \lim_{h \to 0} \frac{\|r(h)\|_W}{\|h\|_V} = 0,$$

where $A = (Df)_a$. Fix $\lambda \in \mathbb{R}$. Then

$$(\lambda f)(a+h) = (\lambda f)(a) + \lambda A(h) + \lambda r(h).$$

Since $\|\lambda r(h)\|_W = |\lambda| \, \|r(h)\|_W$, we have

$$\frac{\|\lambda r(h)\|_W}{\|h\|_V} = |\lambda| \frac{\|r(h)\|_W}{\|h\|_V} \longrightarrow 0.$$

Therefore $\lambda f$ is Fréchet differentiable at $a$ and $(D(\lambda f))_a = \lambda A = \lambda (Df)_a$. □

## Product rule

**Proposition 16** (Product rule)**.** *Let $f, g : \Omega \to \mathbb{R}$ and let $a \in \Omega$. If $f$ and $g$ are Fréchet differentiable at $a$, then the pointwise product $fg : \Omega \to \mathbb{R}$ is Fréchet differentiable at $a$ and*

$$(D(fg))_a = f(a)\,(Dg)_a + g(a)\,(Df)_a,$$

*where the right-hand side is a linear map $V \to \mathbb{R}$ given by*

$$\big(f(a)\,(Dg)_a + g(a)\,(Df)_a\big)(h) = f(a)\,(Dg)_a(h) + g(a)\,(Df)_a(h) \qquad (h \in V).$$

*Proof.*
Let $f, g : \Omega \to \mathbb{R}$ be Fréchet differentiable at $a$. Write

$$f(a+h) = f(a) + A(h) + r_f(h), \qquad \lim_{h\to 0} \frac{|r_f(h)|}{\|h\|_V} = 0,$$

$$g(a+h) = g(a) + B(h) + r_g(h), \qquad \lim_{h\to 0} \frac{|r_g(h)|}{\|h\|_V} = 0,$$

where $A = (Df)_a$ and $B = (Dg)_a$ (bounded linear maps $V \to \mathbb{R}$). For $h$ with $a + h \in \Omega$,

$$\begin{aligned}
(fg)(a+h) &= f(a+h)\,g(a+h) \\
&= \big(f(a) + A(h) + r_f(h)\big)\big(g(a) + B(h) + r_g(h)\big) \\
&= f(a)g(a) + f(a)B(h) + g(a)A(h) \\
&\quad + \underbrace{A(h)B(h)}_{(*)} + \underbrace{f(a)r_g(h) + g(a)r_f(h)}_{(**)} + \underbrace{A(h)r_g(h) + B(h)r_f(h) + r_f(h)r_g(h)}_{(***)}.
\end{aligned}$$

Define the linear map $L : V \to \mathbb{R}$ by

$$L(h) := f(a)B(h) + g(a)A(h).$$

Then

$$(fg)(a+h) = (fg)(a) + L(h) + R(h),$$

where $R(h)$ is the sum of the terms $(*), (**), (***)$ above. We claim that

$$\frac{|R(h)|}{\|h\|_V} \to 0 \qquad (h \to 0).$$

Since $A$ and $B$ are bounded linear maps $V \to \mathbb{R}$, there exist constants $M_A, M_B \geq 0$ such that

$$|A(h)| \leq M_A\|h\|_V, \qquad |B(h)| \leq M_B\|h\|_V \qquad (h \in V).$$

For $(*)$ we have

$$\frac{|A(h)B(h)|}{\|h\|_V} \leq \frac{M_A\|h\|_V\, M_B\|h\|_V}{\|h\|_V} = M_A M_B\|h\|_V \to 0.$$

For $(**)$ we have

$$\frac{|f(a)r_g(h) + g(a)r_f(h)|}{\|h\|_V} \leq |f(a)|\frac{|r_g(h)|}{\|h\|_V} + |g(a)|\frac{|r_f(h)|}{\|h\|_V} \to 0.$$

For $(***)$ we have

$$\frac{|A(h)r_g(h)|}{\|h\|_V} \leq M_A|r_g(h)|, \qquad \frac{|B(h)r_f(h)|}{\|h\|_V} \leq M_B|r_f(h)|.$$

Since $r_f(h) = o(\|h\|_V)$ and $r_g(h) = o(\|h\|_V)$, in particular $r_f(h) \to 0$ and $r_g(h) \to 0$, hence both terms tend to 0. Finally,

$$\frac{|r_f(h) r_g(h)|}{\|h\|_V} = \left( \frac{|r_f(h)|}{\|h\|_V} \right) |r_g(h)| \longrightarrow 0 \cdot 0 = 0.$$

Thus $|R(h)|/\|h\|_V \to 0$, so $fg$ is Fréchet differentiable at $a$ and $(D(fg))_a = L = f(a)(Dg)_a + g(a)(Df)_a$. □

**Proposition 17** (Topological Space Version-A bounded factor times a vanishing factor vanishes)**.** *Let $X$ be a topological space, let $D \subseteq X$, and let $a$ be an accumulation point of $D$. Let $u : D \to \mathbb{R}$ and $v : D \to \mathbb{R}$ be functions such that*

$$\lim_{x \to a} u(x) = 0,$$

*and such that $v$ is bounded in a neighbourhood of $a$, meaning: there exist a neighbourhood $U$ of $a$ in $X$ and a constant $C > 0$ such that*

$$x \in D \cap (U \setminus \{a\}) \quad \Longrightarrow \quad |v(x)| \leq C.$$

*Then*

$$\lim_{x \to a} u(x) v(x) = 0.$$

*Proof.*

Let $\varepsilon > 0$. Choose a neighbourhood $U$ of $a$ and $C > 0$ as in the boundedness assumption for $v$. Since $\lim_{x \to a} u(x) = 0$, there exists a neighbourhood $U_1$ of $a$ such that

$$x \in D \cap (U_1 \setminus \{a\}) \quad \Longrightarrow \quad |u(x)| < \frac{\varepsilon}{C}.$$

Set $U_0 := U \cap U_1$, which is a neighbourhood of $a$. Then for all $x \in D \cap (U_0 \setminus \{a\})$ we have $|v(x)| \leq C$ and $|u(x)| < \varepsilon/C$, hence

$$|u(x)v(x)| = |u(x)|\,|v(x)| < \frac{\varepsilon}{C}\, C = \varepsilon.$$

This is the $\varepsilon$–neighbourhood condition for $\lim_{x \to a} u(x)v(x) = 0$. □

**Proposition 18** (Metric Space Version-A bounded factor times a vanishing factor vanishes)**.** *Let $(X, d)$ be a metric space, let $D \subseteq X$, and let $a$ be an accumulation point of $D$. Let $u : D \to \mathbb{R}$ and $v : D \to \mathbb{R}$ be functions such that*

$$\lim_{x \to a} u(x) = 0,$$

*and such that $v$ is bounded near $a$, meaning: there exist $\delta > 0$ and $C > 0$ such that*

$$x \in D, \quad 0 < d(x, a) < \delta \quad \Longrightarrow \quad |v(x)| \leq C.$$

*Then*

$$\lim_{x \to a} u(x) v(x) = 0.$$

*Proof.*
Let $\varepsilon > 0$. Choose $\delta > 0$ and $C > 0$ as in the boundedness assumption for $v$. Since $\lim_{x\to a} u(x) = 0$, there exists $\eta > 0$ such that for all $x \in D$,

$$0 < d(x,a) < \eta \quad \Longrightarrow \quad |u(x)| < \frac{\varepsilon}{C}.$$

Set $\rho := \min\{\delta, \eta\}$. Then for all $x \in D$ with $0 < d(x,a) < \rho$, we have $|v(x)| \le C$ and $|u(x)| < \varepsilon/C$, hence

$$|u(x)v(x)| = |u(x)|\,|v(x)| < \frac{\varepsilon}{C}\,C = \varepsilon.$$

This is the $\varepsilon$–$\rho$ condition for $\lim_{x\to a} u(x)v(x) = 0$. □

**Proposition 19** (Normed-Space Version-A bounded factor times a vanishing factor vanishes)**.** *Let $(V, \|\cdot\|_V)$ be a normed $\mathbb{R}$-vector space, let $D \subseteq V$, and let $a$ be an accumulation point of $D$ (with respect to the metric induced by $\|\cdot\|_V$). Let $u : D \to \mathbb{R}$ and $v : D \to \mathbb{R}$ be functions such that*

$$\lim_{x\to a} u(x) = 0,$$

*and such that $v$ is bounded near $a$, meaning: there exist $\delta > 0$ and $C > 0$ such that*

$$x \in D, \quad 0 < \|x - a\|_V < \delta \quad \Longrightarrow \quad |v(x)| \le C.$$

*Then*

$$\lim_{x\to a} u(x)v(x) = 0.$$

*Proof.*
Let $\varepsilon > 0$. Choose $\delta > 0$ and $C > 0$ as in the boundedness assumption for $v$. Since $\lim_{x\to a} u(x) = 0$, there exists $\eta > 0$ such that for all $x \in D$,

$$0 < \|x - a\|_V < \eta \quad \Longrightarrow \quad |u(x)| < \frac{\varepsilon}{C}.$$

Set $\rho := \min\{\delta, \eta\}$. Then for all $x \in D$ with $0 < \|x - a\|_V < \rho$, we have $|v(x)| \le C$ and $|u(x)| < \varepsilon/C$, hence

$$|u(x)v(x)| = |u(x)|\,|v(x)| < \frac{\varepsilon}{C}\,C = \varepsilon.$$

This is the $\varepsilon$–$\rho$ condition for $\lim_{x\to a} u(x)v(x) = 0$. □

## Chain rule

**Proposition 20** (Chain rule)**.** *Let $(Z, \|\cdot\|_Z)$ be a normed $\mathbb{R}$-vector space. Let $f : \Omega \to W$ and let $E \subseteq W$ be open with $f(\Omega) \subseteq E$. Let $g : E \to Z$. Fix $a \in \Omega$. If $f$ is Fréchet differentiable at $a$ and $g$ is Fréchet differentiable at $f(a)$, then $g \circ f : \Omega \to Z$ is Fréchet differentiable at $a$ and*

$$(D(g \circ f))_a = (Dg)_{f(a)} \circ (Df)_a,$$

*where the right-hand side is the composition of linear maps $V \to W$ and $W \to Z$.*

*Proof.*
Assume $f$ is Fréchet differentiable at $a$ and $g$ is Fréchet differentiable at $f(a)$. Write

$$f(a+h) = f(a) + A(h) + r_f(h), \qquad \lim_{h\to 0} \frac{\|r_f(h)\|_W}{\|h\|_V} = 0,$$

where $A = (Df)_a$. Also, since $g$ is Fréchet differentiable at $f(a)$, there exist a bounded linear map $B : W \to Z$ and a remainder $r_g : W \to Z$ (defined on a neighbourhood of 0 in $W$) such that for $k \to 0$ in $W$,

$$g(f(a)+k) = g(f(a)) + B(k) + r_g(k), \qquad \lim_{k\to 0} \frac{\|r_g(k)\|_Z}{\|k\|_W} = 0,$$

where $B = (Dg)_{f(a)}$.

For $h$ sufficiently close to 0 we have $a + h \in \Omega$ and thus

$$(g \circ f)(a+h) = g(f(a+h)) = g\big(f(a) + A(h) + r_f(h)\big).$$

Apply the expansion for $g$ with $k = A(h) + r_f(h)$:

$$(g \circ f)(a+h) = g(f(a)) + B(A(h) + r_f(h)) + r_g(A(h) + r_f(h)).$$

Using linearity of $B$ gives

$$(g \circ f)(a+h) = (g \circ f)(a) + (B \circ A)(h) + \underbrace{B(r_f(h)) + r_g(A(h) + r_f(h))}_{=:R(h)}.$$

We show $\|R(h)\|_Z/\|h\|_V \to 0$. Since $B$ is bounded, there exists $M \geq 0$ such that $\|B(w)\|_Z \leq M\|w\|_W$ for all $w \in W$, hence

$$\frac{\|B(r_f(h))\|_Z}{\|h\|_V} \leq M \frac{\|r_f(h)\|_W}{\|h\|_V} \to 0.$$

Next, since $A$ is bounded there exists $M_A \geq 0$ with $\|A(h)\|_W \leq M_A\|h\|_V$, and since $r_f(h) = o(\|h\|_V)$ we have $\|r_f(h)\|_W \to 0$. Therefore $\|A(h)+r_f(h)\|_W \to 0$ as $h \to 0$. Using $\|r_g(k)\|_Z/\|k\|_W \to 0$ as $k \to 0$, we obtain

$$\frac{\|r_g(A(h) + r_f(h))\|_Z}{\|A(h) + r_f(h)\|_W} \to 0.$$

Moreover,

$$\frac{\|A(h) + r_f(h)\|_W}{\|h\|_V} \leq \frac{\|A(h)\|_W}{\|h\|_V} + \frac{\|r_f(h)\|_W}{\|h\|_V} \leq M_A + \frac{\|r_f(h)\|_W}{\|h\|_V},$$

so the ratio $\|A(h) + r_f(h)\|_W/\|h\|_V$ is bounded for $h$ near 0. Hence, by proposition 18,

$$\frac{\|r_g(A(h) + r_f(h))\|_Z}{\|h\|_V} = \left(\frac{\|r_g(A(h) + r_f(h))\|_Z}{\|A(h) + r_f(h)\|_W}\right)\left(\frac{\|A(h) + r_f(h)\|_W}{\|h\|_V}\right) \to 0.$$

Therefore $\|R(h)\|_Z/\|h\|_V \to 0$, and $(g \circ f)$ is Fréchet differentiable at $a$ with

$$(D(g \circ f))_a = B \circ A = (Dg)_{f(a)} \circ (Df)_a.$$

□

**Remark 13** (Interpretation)**.** *The chain rule states that the first-order (linear) approximation of the composite map $g \circ f$ at $a$ is obtained by composing the first-order approximations of $f$ at $a$ and $g$ at $f(a)$.*

## Quotient rule

**Proposition 21** (Quotient rule)**.** *Let $f, g : \Omega \to \mathbb{R}$ and let $a \in \Omega$ with $g(a) \neq 0$. Assume $f$ and $g$ are Fréchet differentiable at $a$. Then there exists an open neighbourhood $U \subseteq \Omega$ of $a$ such that $g(x) \neq 0$ for all $x \in U$, hence the quotient*

$$\frac{f}{g} : U \to \mathbb{R}, \qquad x \mapsto \frac{f(x)}{g(x)},$$

*is well-defined. Moreover, $f/g$ is Fréchet differentiable at $a$ and*

$$\left(D\left(\frac{f}{g}\right)\right)_a = \frac{g(a)\,(Df)_a - f(a)\,(Dg)_a}{(g(a))^2}.$$

*Proof.*
Let $f, g : \Omega \to \mathbb{R}$ be Fréchet differentiable at $a$ with $g(a) \neq 0$. Since Fréchet differentiability implies continuity, $g$ is continuous at $a$ and therefore there exists an open neighbourhood $U \subseteq \Omega$ of $a$ such that $g(x) \neq 0$ for all $x \in U$.

Define $h : (\mathbb{R} \setminus \{0\}) \to \mathbb{R}$ by $h(t) = 1/t$. The map $h$ is differentiable at $g(a)$ with

$$(Dh)_{g(a)}(s) = -\frac{s}{(g(a))^2} \qquad (s \in \mathbb{R}),$$

i.e. $(Dh)_{g(a)} = -\frac{1}{(g(a))^2}\,\mathrm{id}_{\mathbb{R}}$. By the chain rule, the composition $h \circ g : U \to \mathbb{R}$ is Fréchet differentiable at $a$ and

$$(D(h \circ g))_a = (Dh)_{g(a)} \circ (Dg)_a = -\frac{1}{(g(a))^2}(Dg)_a.$$

On $U$ we have $h \circ g = 1/g$, hence $1/g$ is Fréchet differentiable at $a$ and

$$\left(D\left(\frac{1}{g}\right)\right)_a = -\frac{1}{(g(a))^2}(Dg)_a.$$

Now on $U$ we can write

$$\frac{f}{g} = f \cdot \frac{1}{g}.$$

Applying the product rule (already proved) to the real-valued functions $f$ and $1/g$ yields

$$\left(D\left(\frac{f}{g}\right)\right)_a = f(a)\left(D\left(\frac{1}{g}\right)\right)_a + \frac{1}{g(a)}(Df)_a.$$

Substituting $(D(1/g))_a = -\frac{1}{(g(a))^2}(Dg)_a$ gives

$$\left(D\left(\frac{f}{g}\right)\right)_a = f(a)\left(-\frac{1}{(g(a))^2}(Dg)_a\right) + \frac{1}{g(a)}(Df)_a = \frac{g(a)\,(Df)_a - f(a)\,(Dg)_a}{(g(a))^2}.$$

□

**Remark 14** (Why $f/g$ is defined near $a$)**.** *Since $g$ is continuous at $a$ (Fréchet differentiability implies continuity) and $g(a) \neq 0$, there exists an open neighbourhood $U \subseteq \Omega$ of $a$ such that $g(x) \neq 0$ for all $x \in U$.*

# Graphs, Level Sets, Gradients, and Tangent Hyperplanes

## Standing conventions

Throughout, $\mathbb{F}$ denotes $\mathbb{R}$. Let $(V, \|\cdot\|_V)$ be a normed $\mathbb{R}$-vector space, let $\Omega \subseteq V$ be open, and let $f : \Omega \to \mathbb{R}$ be a map. If $f$ is Fréchet differentiable at $a \in \Omega$, then its derivative is denoted by $(Df)_a : V \to \mathbb{R}$. In the Euclidean case $V = \mathbb{R}^d$, we use the standard inner product $\langle x, y\rangle = \sum_{i=1}^d x_i y_i$ and the corresponding Euclidean norm $\|\cdot\|_2$.

## Graph of a function

**Definition 10** (Graph of a function)**.** *Let $f : \Omega \to \mathbb{R}$. The* graph *of $f$ is the subset of $V \times \mathbb{R}$ defined by*

$$\operatorname{graph}(f) := \{(x, t) \in V \times \mathbb{R} : x \in \Omega,\ t = f(x)\}.$$

## Level set

**Definition 11** (Level set)**.** *Let $c \in \mathbb{R}$. The* level set *of $f$ at level $c$ is the subset of $\Omega$ defined by*

$$L_c := \{x \in \Omega : f(x) = c\} = f^{-1}(\{c\}).$$

*If $a \in \Omega$, the* level set through $a$ *is $L_{f(a)}$.*

## Tangent hyperplane to the graph

**Definition 12** (Tangent hyperplane to the graph)**.** *Let $a \in \Omega$ and set $p := (a, f(a)) \in V \times \mathbb{R}$. Assume $f$ is Fréchet differentiable at $a$. The* tangent hyperplane to $\operatorname{graph}(f)$ at $p$ *is the affine subset of $V \times \mathbb{R}$ defined by*

$$T_p \operatorname{graph}(f) := \{(x, t) \in V \times \mathbb{R} : t = f(a) + (Df)_a(x - a)\}.$$

**Remark 15** (Interpretation)**.** *The tangent hyperplane to the graph is the graph of the affine map*

$$x \longmapsto f(a) + (Df)_a(x - a),$$

*which is the first-order (affine) approximation of $f$ at $a$.*

## Tangent hyperplane to a level set

**Definition 13** (Tangent hyperplane to a level set)**.** *Let $a \in \Omega$ and set $c := f(a)$, so that $a \in L_c$. Assume $f$ is Fréchet differentiable at $a$. The* tangent hyperplane to the level set $L_c$ at $a$ *is the affine subset of $V$ defined by*

$$T_a L_c := \{x \in V : (Df)_a(x - a) = 0\} = a + \ker\big((Df)_a\big).$$

**Remark 16** (Interpretation)**.** *Fix $a \in \Omega$ and set $c := f(a)$, so the level set through $a$ is*

$$L_c = \{x \in \Omega : f(x) = c\}.$$

*A point $x \in L_c$ satisfies the constraint $f(x) = f(a)$, equivalently $f(x) - f(a) = 0$. Writing $x = a + h$, this becomes*

$$f(a+h) - f(a) = 0.$$

*Since $f$ is Fréchet differentiable at $a$, there exists a remainder $r(h)$ such that*

$$f(a+h) - f(a) = (Df)_a(h) + r(h), \qquad \frac{|r(h)|}{\|h\|_V} \to 0 \quad (h \to 0).$$

*Thus, for small $h$, the quantity $f(a+h) - f(a)$ is well-approximated by the linear term $(Df)_a(h)$, and the remainder $r(h)$ is of smaller order than $\|h\|_V$. The equation $f(a+h) - f(a) = 0$ can therefore be viewed as*

$$(Df)_a(h) + r(h) = 0,$$

*and its* linearisation at *$a$ is obtained by discarding the higher-order term $r(h)$, yielding the linear condition*

$$(Df)_a(h) = 0.$$

*The linearised constraint is $(Df)_a(h) = 0$, whose solution set in the increment variable $h$ is*

$$\ker((Df)_a) = \{h \in V : (Df)_a(h) = 0\}.$$

*A point $x \in V$ lies on the tangent hyperplane to the level set $L_c$ at $a$ precisely when its increment from $a$, $h = x - a$, belongs to this kernel. Thus the tangent hyperplane is*

$$\{x \in V : x - a \in \ker((Df)_a)\} = \{x \in V : (Df)_a(x-a) = 0\} = a + \ker((Df)_a).$$

*It consists of exactly those directions $h$ at $a$ for which the first-order change of $f$ vanishes, in the sense that $(Df)_a(h) = 0$ implies*

$$f(a+h) = f(a) + o(\|h\|_V) \qquad (h \to 0 \text{ along the direction } h).$$

## Gradient in $\mathbb{R}^d$

**Definition 14** (Gradient)**.** *Assume $V = \mathbb{R}^d$ with the standard inner product $\cdot$. Let $f : \Omega \subseteq \mathbb{R}^d \to \mathbb{R}$ be Fréchet differentiable at $a \in \Omega$. The* gradient of $f$ at *$a$ is the unique vector $\nabla f(a) \in \mathbb{R}^d$ such that*

$$(Df)_a(h) = \langle \nabla f(a), h \rangle \qquad (h \in \mathbb{R}^d).$$

**Remark 17** (Coordinate expression)**.** *If $f$ has partial derivatives at $a$ and is Fréchet differentiable at $a$, then*

$$\nabla f(a) = \big(\partial_1 f(a), \ldots, \partial_d f(a)\big).$$

**Remark 18** (Level-set tangent hyperplane in terms of the gradient)**.** *Assume $V = \mathbb{R}^d$ and $f$ is Fréchet differentiable at $a$. Then the tangent hyperplane to the level set $L_{f(a)}$ at $a$ can be written as*

$$T_a L_{f(a)} = \{x \in \mathbb{R}^d : \langle \nabla f(a), (x-a) \rangle = 0\}.$$

*In particular, $\nabla f(a)$ is normal to $T_a L_{f(a)}$.*

# Compactness and Attainment of Extrema

## Standing conventions

Throughout, $(V, \|\cdot\|_V)$ denotes a finite-dimensional normed $\mathbb{R}$-vector space. All topological notions (open/closed/compact/continuous) are with respect to the metric induced by the norm.

## Finite-dimensional spaces are not compact

**Proposition 22** (A normed vector space is bounded only in the trivial case)**.** *Let $(V, \|\cdot\|_V)$ be a normed $\mathbb{R}$-vector space. Then $V$ is bounded as a subset of itself (with respect to the metric induced by $\|\cdot\|_V$) if and only if $V = \{0\}$.*

*Proof.*
If $V = \{0\}$, then $\|x\|_V = 0$ for all $x \in V$, hence $V$ is bounded.

Conversely, assume $V \neq \{0\}$ and choose $v \in V$ with $v \neq 0$. For each $t \in \mathbb{R}$, the vector $tv$ belongs to $V$, and by absolute homogeneity of the norm,

$$\|tv\|_V = |t|\,\|v\|_V.$$

As $t \to \infty$, the right-hand side tends to $\infty$, so the set $\{\|x\|_V : x \in V\}$ is unbounded in $\mathbb{R}$. Equivalently, there is no constant $M > 0$ such that $\|x\|_V \leq M$ for all $x \in V$. Thus $V$ is not bounded. □

**Remark 19** (Compactness of a finite-dimensional normed space)**.** *Compactness is a property of* topological spaces*. Since a normed vector space $(V, \|\cdot\|_V)$ carries the topology induced by its norm, it makes sense to ask whether the underlying topological space $V$ is compact. If $V$ is finite-dimensional, then by Heine–Borel a subset $K \subseteq V$ is compact if and only if it is closed and bounded. Applying this with $K = V$ (which is closed in itself) shows that $V$ is compact if and only if $V$ is bounded as a subset of itself. By Proposition 21 this occurs only in the trivial case $V = \{0\}$.*

## Compact subsets in finite dimensions

**Proposition 23** (Heine–Borel in finite-dimensional normed spaces)**.** *Let $(V, \|\cdot\|_V)$ be a* finite-dimensional *normed $\mathbb{R}$-vector space, and let $K \subseteq V$. Then $K$ is compact if and only if $K$ is closed and bounded.*

**Remark 20** (Important examples)**.** *Let $(V, \|\cdot\|_V)$ be a finite-dimensional normed $\mathbb{R}$-vector space. The* closed unit ball *and* unit sphere *are the subsets*

$$\overline{B}(0,1) := \{v \in V : \|v\|_V \leq 1\}, \qquad S := \{v \in V : \|v\|_V = 1\}.$$

*Both sets are bounded. Moreover, both are closed. Hence $\overline{B}(0,1)$ and $S$ are compact by the Heine–Borel proposition above.*

## Extrema on compact sets

**Proposition 24** (Extreme value theorem)**.** *Let $K$ be a compact topological space and let $f : K \to \mathbb{R}$ be continuous. Then there exist points $x_{\min}, x_{\max} \in K$ such that*

$$f(x_{\min}) = \min_{x \in K} f(x), \qquad f(x_{\max}) = \max_{x \in K} f(x).$$

**Remark 21** (Continuity is essential)**.** *The conclusion of the extreme value theorem can fail if $f$ is not continuous, even when $K$ is compact.*

# The Operator Norm

## Standing conventions

Throughout, $(V, \|\cdot\|_V)$ and $(W, \|\cdot\|_W)$ denote normed $\mathbb{R}$-vector spaces. We write $B(V, W)$ for the vector space of bounded linear maps $V \to W$. If $L \in B(V, W)$, then $\|L(v)\|_W$ denotes the norm of the vector $L(v) \in W$ (not a norm of $L$). All suprema below are taken in $\mathbb{R}$.

## Upper bounds and supremum

**Definition 15** (Upper bound and supremum)**.** *Let $S \subseteq \mathbb{R}$. A number $M \in \mathbb{R}$ is an* upper bound *for $S$ if $s \leq M$ for all $s \in S$. If $S$ is bounded above, its* supremum *is the real number* $\sup S$ *characterised by:*

1. $\sup S$ *is an upper bound for $S$, and*
2. *if $U$ is any upper bound for $S$, then* $\sup S \leq U$.

## Definition of the operator norm

**Definition 16** (Operator norm)**.** *Let $L \in B(V, W)$. The* operator norm *of $L$ is*

$$\|L\|_{V\to W} := \sup\{\|L(x)\|_W : \ x \in V, \ \|x\|_V \leq 1\}.$$

**Remark 22** (Finite-dimensional case)**.** *If $V$ is finite-dimensional, then the closed unit ball $\{x \in V : \|x\|_V \leq 1\}$ is compact and the map $x \mapsto \|L(x)\|_W$ is continuous, hence the supremum in the definition is attained. Therefore*

$$\|L\|_{V\to W} = \max\{\|L(x)\|_W : \ x \in V, \ \|x\|_V \leq 1\}.$$

**Proposition 25** (Operator norm as a least constant)**.** *Let $L \in B(V, W)$. Then*

$$\|L\|_{V\to W} = \min\Big\{ C \in [0, \infty) : \|L(v)\|_W \leq C\,\|v\|_V \text{ for all } v \in V \Big\}.$$

**Proposition 26** (Equivalent ratio formula)**.** *Let $L \in B(V, W)$. Then*

$$\|L\|_{V\to W} = \sup\left\{ \frac{\|L(v)\|_W}{\|v\|_V} : \ v \in V, \ v \neq 0 \right\}.$$

*Proof.*
Let $v \neq 0$ and set $u := v/\|v\|_V$, so that $\|u\|_V = 1$. By linearity,

$$L(v) = L(\|v\|_V u) = \|v\|_V\, L(u),$$

so

$$\frac{\|L(v)\|_W}{\|v\|_V} = \|L(u)\|_W.$$

Hence the set $\{\|L(u)\|_W : \ \|u\|_V = 1\}$ coincides with the set $\{\|L(v)\|_W/\|v\|_V : \ v \neq 0\}$. Taking suprema and using $\sup_{\|x\|_V \leq 1} \|L(x)\|_W = \sup_{\|x\|_V = 1} \|L(x)\|_W$ gives the claim. $\square$

**Remark 23.** *The "worst" stretching occurs on the boundary of the unit ball*

**Remark 24** (Why the definition uses the closed unit ball)**.** *Proposition 25 shows that the quantity*

$$\sup\left\{\frac{\|L(v)\|_W}{\|v\|_V} : v \in V,\ v \neq 0\right\}$$

*can be computed by restricting to vectors of norm* 1*, because every* $v \neq 0$ *can be written as* $v = \|v\|_V\, u$ *with* $\|u\|_V = 1$ *and then* $\|L(v)\|_W/\|v\|_V = \|L(u)\|_W$*. Since the unit sphere* $\{u \in V : \|u\|_V = 1\}$ *is contained in the closed unit ball* $\{x \in V : \|x\|_V \leq 1\}$*, one may equivalently define*

$$\|L\|_{V\to W} = \sup\{\|L(x)\|_W :\ \|x\|_V \leq 1\}.$$

*Thus the use of the closed unit ball in the definition is a convenient normalisation: it collects all directions with input size at most* 1 *into a single set, and it yields the same least upper bound as the ratio formula.*

**Remark 25** (Conceptual meaning and relation between the two characterisations)**.** *Let* $L \in B(V, W)$*. By definition,*

$$\|L\|_{V\to W} = \sup\{\|L(x)\|_W :\ x \in V,\ \|x\|_V \leq 1\},$$

*so* $\|L\|_{V\to W}$ *is the least upper bound on the set of output norms produced by inputs of norm at most* 1*.*

*For each nonzero* $v \in V$*, the ratio*

$$\frac{\|L(v)\|_W}{\|v\|_V}$$

*measures the norm expansion factor of* $L$ *in the direction* $v$*: it compares the size of the output vector* $L(v)$ *to the size of the input vector* $v$*. The operator norm* $\|L\|_{V\to W}$ *is the least upper bound of these expansion factors, that is,*

$$\frac{\|L(v)\|_W}{\|v\|_V} \leq \|L\|_{V\to W} \qquad (v \in V,\ v \neq 0).$$

*Equivalently,* $\|L\|_{V\to W}$ *is the smallest constant* $C \geq 0$ *such that*

$$\|L(v)\|_W \leq C\,\|v\|_V \qquad (v \in V),$$

*which is the content of the least characterisation in the preceding proposition.*

*In this sense,* $\|L\|_{V\to W}$ *is the maximal stretching factor of* $L$ *with respect to the given norms: no vector is stretched by a factor larger than* $\|L\|_{V\to W}$*, and* $\|L\|_{V\to W}$ *is the least number with this property.*

**Remark 26** (Why one may write min in finite dimensions)**.** *Let* $(V, \|\cdot\|_V)$ *be finite-dimensional and let* $L \in B(V, W)$*. Consider the set of constants*

$$\mathcal{C} := \Big\{\, C \in [0, \infty) : \|L(v)\|_W \leq C\,\|v\|_V \text{ for all } v \in V \,\Big\}.$$

*In general normed spaces, one always has*

$$\|L\|_{V\to W} = \inf \mathcal{C},$$

*because $\|L\|_{V\to W}$ is the least upper bound of the stretching factors and hence the least constant for which the estimate $\|L(v)\|_W \le C\|v\|_V$ holds for all $v$.*

*In finite dimensions, the unit sphere*

$$S := \{v \in V : \|v\|_V = 1\}$$

*is compact (Heine–Borel), and the function $\phi : S \to \mathbb{R}$ defined by*

$$\phi(v) := \|L(v)\|_W$$

*is continuous. By the extreme value theorem, $\phi$ attains a maximum at some $v_0 \in S$. Setting*

$$C_0 := \phi(v_0) = \|L(v_0)\|_W,$$

*one has $\|L(v)\|_W \le C_0\|v\|_V$ for all $v \in V$, and no smaller constant works because equality holds for $v = v_0$. Hence the infimum of $\mathcal{C}$ is attained, and therefore*

$$\|L\|_{V\to W} = \min \mathcal{C}.$$

## Basic inequality for the operator norm

**Proposition 27** (Basic inequality for the operator norm)**.** *Let $L \in B(V, W)$. Then for every $v \in V$,*

$$\|L(v)\|_W \le \|L\|_{V\to W}\, \|v\|_V.$$

*Proof.*
If $v = 0$, then $\|L(v)\|_W = 0$ and the inequality holds. Assume $v \neq 0$ and set $u := v/\|v\|_V$, so that $\|u\|_V = 1$. Then $v = \|v\|_V u$ and by linearity

$$L(v) = L(\|v\|_V u) = \|v\|_V\, L(u),$$

hence

$$\|L(v)\|_W = \|v\|_V\, \|L(u)\|_W.$$

Since $\|u\|_V \le 1$, the value $\|L(u)\|_W$ belongs to the set $\{\|L(x)\|_W : \ \|x\|_V \le 1\}$, hence by the defining property of supremum,

$$\|L(u)\|_W \le \sup\{\|L(x)\|_W : \ \|x\|_V \le 1\} = \|L\|_{V\to W}.$$

Multiplying by $\|v\|_V$ gives $\|L(v)\|_W \le \|L\|_{V\to W}\, \|v\|_V$. □

**Remark 27** (Interpretation)**.** *Proposition 26 states that the operator norm $\|L\|_{V\to W}$ is an upper bound on all norm expansion factors of $L$: for every nonzero $v \in V$,*

$$\frac{\|L(v)\|_W}{\|v\|_V} \le \|L\|_{V\to W}.$$

*Equivalently, Proposition 26 states that the same constant $\|L\|_{V\to W}$ bounds $\|L(v)\|_W$ for every input vector $v \in V$, namely*

$$\|L(v)\|_W \le \|L\|_{V\to W}\, \|v\|_V \qquad (v \in V).$$

*(the output norm cannot exceed (maximum stretch factor)×(input norm))*

# The Mean Value Inequality

## Standing conventions

Let $(V, \|\cdot\|_V)$ and $(W, \|\cdot\|_W)$ be normed $\mathbb{R}$-vector spaces. All limits in $W$ are taken with respect to $\|\cdot\|_W$, and all suprema below are taken in $\mathbb{R}$.

If $I \subseteq \mathbb{R}$ is an open interval and $g : I \to W$ is differentiable, then $g'(t) \in W$ denotes its one-variable derivative at $t \in I$.

If $\Omega \subseteq V$ is open and $f : \Omega \to W$ is Fréchet differentiable at $x \in \Omega$, then its derivative is denoted by

$$(Df)_x \in B(V, W),$$

and its operator norm by $\|(Df)_x\|_{V\to W}$.

## The mean value inequality in one variable

**Proposition 28** (Mean value inequality)**.** *Let $[a, b] \subseteq \mathbb{R}$ with $a < b$, and let $g : [a, b] \to W$. Assume that $g$ is continuous on $[a, b]$ and differentiable on $(a, b)$. Assume moreover that the set*

$$S := \{\|g'(t)\|_W : t \in (a, b)\}$$

*is bounded above in $\mathbb{R}$. Then*

$$\|g(b) - g(a)\|_W \leq (b - a) \sup_{t\in(a,b)} \|g'(t)\|_W.$$

*Equivalently, if $M \geq 0$ is any number satisfying*

$$\|g'(t)\|_W \leq M \qquad \text{for all } t \in (a, b),$$

*then*

$$\|g(b) - g(a)\|_W \leq M\,(b - a).$$

**Remark 28** (Interpretation)**.** *The quantity $\|g'(t)\|_W$ is the speed of the curve $t \mapsto g(t)$ in the normed space $W$. If the speed is bounded above on $(a, b)$, then the total displacement from $g(a)$ to $g(b)$ cannot exceed*

$$\textit{(maximal speed)} \times \textit{(time interval length)}.$$

**Remark 29** (Why a supremum appears)**.** *The number $\sup_{t\in(a,b)} \|g'(t)\|_W$ is the least upper bound of the set $S = \{\|g'(t)\|_W : t \in (a, b)\}$. Any upper bound $M$ for $S$ also yields a valid estimate, and one has*

$$\sup_{t\in(a,b)} \|g'(t)\|_W \leq M.$$

*Thus using the supremum gives the sharpest constant available from this information.*

## Line segments and a reduction to one variable

**Definition 17** (Line segment between two points)**.** *Let $a, b \in V$. The (closed) line segment from $a$ to $b$ is*

$$[a, b] := \{(1 - t)a + tb : \ t \in [0, 1]\}.$$

**Definition 18** (Parametrisation of the segment)**.** *Let $a, b \in V$ and define*

$$\gamma : [0, 1] \to V, \qquad \gamma(t) := (1 - t)a + tb.$$

*Then $\gamma([0, 1]) = [a, b]$ and $\gamma$ is differentiable on $(0, 1)$ with*

$$\gamma'(t) = b - a \qquad (t \in (0, 1)).$$

**Remark 30** (What it means to parametrise the segment)**.** *To parametrise a subset $S \subseteq V$ means to describe its points as the image of a map from a simple parameter set. Concretely, it means to exhibit a map $\gamma : D \to V$ from a parameter set $D \subseteq \mathbb{R}$ such that*

$$\gamma(D) = S.$$

*In particular, $\gamma$ describes every point of $S$ as $\gamma(t)$ for some parameter value $t$, and conversely every $\gamma(t)$ lies in $S$. For the line segment,*

$$[a, b] = \{(1 - t)a + tb : \ t \in [0, 1]\},$$

*the map $\gamma : [0, 1] \to V$ given by $\gamma(t) = (1 - t)a + tb$ satisfies $\gamma([0, 1]) = [a, b]$, with $\gamma(0) = a$ and $\gamma(1) = b$.*

**Remark 31** (Reduction to a one-variable map)**.** *Assume $[a, b] \subseteq \Omega$ and define*

$$g : [0, 1] \to W, \qquad g(t) := f(\gamma(t)).$$

*Then*

$$g(0) = f(a), \qquad g(1) = f(b),$$

*so estimating $\|f(b) - f(a)\|_W$ is equivalent to estimating $\|g(1) - g(0)\|_W$.*

**Remark 32** (Idea of the reduction)**.** *The reduction is that the change from $a$ to $b$ in the domain $V$ is described by a one-variable path $\gamma : [0, 1] \to V$ with $\gamma(0) = a$ and $\gamma(1) = b$. Composing with $f$ gives a one-variable map $g = f \circ \gamma : [0, 1] \to W$ with $g(0) = f(a)$ and $g(1) = f(b)$, so*

$$\|f(b) - f(a)\|_W = \|g(1) - g(0)\|_W.$$

## Mean value inequality along a segment

**Proposition 29** (Mean value inequality along a line segment)**.** *Let $a, b \in \Omega$ and assume $[a, b] \subseteq \Omega$. Assume that $f$ is Fréchet differentiable at every point of $[a, b]$. Assume that the set*

$$\left\{ \ \|(Df)_x\|_{V \to W} : \ x \in [a, b] \ \right\}$$

*is bounded above in $\mathbb{R}$. Then*

$$\|f(b) - f(a)\|_W \leq \left( \sup_{x \in [a,b]} \|(Df)_x\|_{V \to W} \right) \|b - a\|_V.$$

*Equivalently, if $M \geq 0$ satisfies $\|(Df)_x\|_{V \to W} \leq M$ for all $x \in [a, b]$, then*

$$\|f(b) - f(a)\|_W \leq M \, \|b - a\|_V.$$

*Proof.*

Define $\gamma(t) = (1 - t)a + tb$ and $g(t) = f(\gamma(t))$ as above. Then $g(0) = f(a)$ and $g(1) = f(b)$. Since $f$ is Fréchet differentiable on $[a, b]$, the chain rule gives that $g$ is differentiable on $(0, 1)$ and

$$g'(t) = (Df)_{\gamma(t)}(\gamma'(t)) \qquad (t \in (0, 1)).$$

Because $\gamma'(t) = b - a$, we have

$$g'(t) = (Df)_{\gamma(t)}(b - a) \qquad (t \in (0, 1)).$$

Taking norms and using the basic operator norm inequality $\|L(v)\|_W \le \|L\|_{V\to W}\|v\|_V$ yields

$$\|g'(t)\|_W = \|(Df)_{\gamma(t)}(b-a)\|_W \le \|(Df)_{\gamma(t)}\|_{V\to W}\,\|b-a\|_V \qquad (t \in (0,1)).$$

Set

$$M := \sup_{x\in[a,b]} \|(Df)_x\|_{V\to W}.$$

Then $\|(Df)_{\gamma(t)}\|_{V\to W} \le M$ for all $t \in (0,1)$, hence

$$\|g'(t)\|_W \le M\,\|b-a\|_V \qquad (t \in (0,1)).$$

Applying the one-variable mean value inequality to $g : [0,1] \to W$ gives

$$\|g(1) - g(0)\|_W \le M\,\|b-a\|_V.$$

Since $g(1) - g(0) = f(b) - f(a)$, this is the desired inequality. □

# Multilinear Maps and Higher Fréchet Derivatives

## Standing conventions

Let $(V, \|\cdot\|_V)$ and $(W, \|\cdot\|_W)$ be normed $\mathbb{R}$-vector spaces and let $\Omega \subseteq V$ be open. We write $B(V,W)$ for the vector space of bounded linear maps $V \to W$ (with the operator norm).

**Definition 19** ($k$-linear (multilinear) map)**.** *Let $k \in \mathbb{N}$ and let $V_1, \dots, V_k$ and $W$ be $\mathbb{R}$-vector spaces. A map*

$$T : V_1 \times \cdots \times V_k \to W$$

*is called* $k$-linear *(or* multilinear*) if, for each $i \in \{1, \dots, k\}$, the map*

$$V_i \to W, \qquad x \mapsto T(v_1, \dots, v_{i-1}, x, v_{i+1}, \dots, v_k),$$

*is linear for every fixed choice of $v_j \in V_j$ with $j \neq i$.*

## First derivative: pointwise derivative and derivative map

**Remark 33** (Pointwise derivative)**.** *Let $f : \Omega \to W$ and let $a \in \Omega$. If $f$ is Fréchet differentiable at $a$, then its pointwise derivative is a bounded linear map*

$$(Df)_a : V \to W.$$

*The input of $(Df)_a$ is an* increment *$h \in V$ (a direction/step from the base point $a$), and $(Df)_a(h)$ is the linear approximation to the change $f(a+h) - f(a)$ for small $h$. The condition $a + h \in \Omega$ is only needed to ensure that $f(a+h)$ is defined; it does not change the fact that $(Df)_a$ is a linear map on $V$.*

**Remark 34** (Derivative map)**.** *Assume $f$ is Fréchet differentiable at every point of $\Omega$. The* derivative map *is the function*

$$Df : \Omega \to B(V,W), \qquad a \mapsto (Df)_a.$$

*Thus $Df$ takes a base point $a \in \Omega$ and returns the linearisation $(Df)_a$ of $f$ at $a$.*

## Second derivative: three related objects

**Definition 20** (Twice Fréchet differentiable at a point)**.** *Let $f : \Omega \to W$ be Fréchet differentiable on $\Omega$ and let $a \in \Omega$. We say that $f$ is* twice Fréchet differentiable at *$a$ if the derivative map*

$$Df : \Omega \to B(V, W)$$

*is Fréchet differentiable at $a$ (as a map into the normed space $B(V, W)$).*

**Remark 35** (Second derivative map)**.** *If $f$ is twice Fréchet differentiable at every point of $\Omega$, then $Df$ is Fréchet differentiable on $\Omega$ and its derivative map is*

$$D(Df) : \Omega \to B\big(V, B(V, W)\big), \qquad a \mapsto (D(Df))_a.$$

*Thus $D(Df)$ takes a base point $a \in \Omega$ and returns the pointwise derivative $(D(Df))_a$.*

**Remark 36** (Second pointwise derivative)**.** *If $f$ is twice Fréchet differentiable at $a$, then the pointwise derivative of $Df$ at $a$ is a bounded linear map*

$$(D(Df))_a : V \to B(V, W).$$

*Its input is an increment $h \in V$ (moving the base point from $a$ to $a + h$), and its output $(D(Df))_a(h)$ is a bounded linear map $V \to W$.*

**Definition 21** (Second derivative as a bilinear map)**.** *Assume $f$ is twice Fréchet differentiable at $a$. Define*

$$(D^2 f)_a : V \times V \to W$$

*by*

$$(D^2 f)_a(h, k) := \big((D(Df))_a(h)\big)(k) \qquad (h, k \in V).$$

**Proposition 30** (Relation between $(D(Df))_a$ and $(D^2 f)_a$)**.** *Assume $f$ is twice Fréchet differentiable at $a$. Then $(D^2 f)_a : V \times V \to W$ is bilinear, and for all $h, k \in V$,*

$$(D^2 f)_a(h, k) = \big((D(Df))_a(h)\big)(k).$$

*Conversely, for each $h \in V$,*

$$(D(Df))_a(h) : V \to W, \qquad k \mapsto (D^2 f)_a(h, k).$$

*Proof.*

Fix $h \in V$. The map $(D(Df))_a(h) \in B(V, W)$ is linear in $k$, hence $k \mapsto (D^2 f)_a(h, k)$ is linear. Fix $k \in V$. Since $(D(Df))_a : V \to B(V, W)$ is linear and evaluation at $k$ is linear, the map $h \mapsto (D^2 f)_a(h, k)$ is linear. Thus $(D^2 f)_a$ is bilinear. The displayed identities follow directly from the definition of $(D^2 f)_a$. □

## Higher derivatives

**Definition 22** (Bounded $n$-linear maps)**.** *Let $n \in \mathbb{N}$. A map $T : V^n \to W$ is called* bounded $n$-linear *if it is $n$-linear and there exists $C \geq 0$ such that*

$$\|T(v_1, \ldots, v_n)\|_W \leq C \, \|v_1\|_V \cdots \|v_n\|_V \qquad (v_1, \ldots, v_n \in V).$$

*The vector space of bounded $n$-linear maps $V^n \to W$ is denoted by $B_n(V, W)$.*

**Definition 23** ($n$-times Fréchet differentiable at a point)**.** *Let $n \in \mathbb{N}$ and let $f : \Omega \to W$. We define inductively what it means for $f$ to be $n$-times Fréchet differentiable at $a \in \Omega$. For $n = 1$ this is the usual Fréchet differentiability. For $n \geq 2$, we say that $f$ is* $n$-times Fréchet differentiable at *$a$ if $f$ is $(n-1)$-times Fréchet differentiable on a neighbourhood of $a$ and the $(n-1)$-st derivative map*

$$D^{n-1}f : \Omega \to B_{n-1}(V, W), \qquad x \mapsto (D^{n-1}f)_x,$$

*is Fréchet differentiable at $a$ (as a map into the normed space $B_{n-1}(V, W)$).*

**Remark 37** (The three objects for the $n$-th derivative)**.** *Assume $f$ is $n$-times Fréchet differentiable on $\Omega$, with $n \geq 2$. Then the following objects are defined:*

1. *The* $n$-th derivative map *(an assignment of $n$-th derivatives to base points)*

$$D^n f : \Omega \to B_n(V, W), \qquad a \mapsto (D^n f)_a.$$

2. *The* $n$-th pointwise derivative *of the map $D^{n-1}f$ at $a$,*

$$\big(D(D^{n-1}f)\big)_a : V \to B_{n-1}(V, W).$$

3. *The* $n$-th derivative at *$a$ as an $n$-linear map,*

$$(D^n f)_a : V^n \to W,$$

*defined from the pointwise derivative by*

$$(D^n f)_a(h_1, \ldots, h_n) = \Big(\big(D(D^{n-1}f)\big)_a(h_1)\Big)(h_2, \ldots, h_n) \qquad (h_1, \ldots, h_n \in V).$$

**Proposition 31** (Multilinearity of $(D^n f)_a$)**.** *Let $n \geq 2$ and assume $f$ is $n$-times Fréchet differentiable at $a$. Then $(D^n f)_a : V^n \to W$ defined above is $n$-linear. Moreover, for each $h_1 \in V$,*

$$\big(D(D^{n-1}f)\big)_a(h_1) : V^{n-1} \to W, \qquad (h_2, \ldots, h_n) \mapsto (D^n f)_a(h_1, \ldots, h_n).$$

*Proof.*
Fix $h_1 \in V$. Then $\big(D(D^{n-1}f)\big)_a(h_1) \in B_{n-1}(V, W)$ is multilinear in $(h_2, \ldots, h_n)$, hence the map $(h_2, \ldots, h_n) \mapsto (D^n f)_a(h_1, \ldots, h_n)$ is multilinear. Next, for fixed $(h_2, \ldots, h_n)$, the map $h_1 \mapsto \big(D(D^{n-1}f)\big)_a(h_1)$ is linear in $B_{n-1}(V, W)$, and evaluation at $(h_2, \ldots, h_n)$ is linear, so $h_1 \mapsto (D^n f)_a(h_1, \ldots, h_n)$ is linear. Therefore $(D^n f)_a$ is $n$-linear, and the displayed identity follows from the definition. □

# Higher-Order Directional Derivatives

## Standing conventions

Let $(V, \|\cdot\|_V)$ and $(W, \|\cdot\|_W)$ be normed $\mathbb{R}$-vector spaces, let $\Omega \subseteq V$ be open, and let $f : \Omega \to W$ be a map. Fix $a \in \Omega$. Directional derivatives are taken in the sense defined earlier:

$$(D_v f)_a = \lim_{t \to 0} \frac{f(a + tv) - f(a)}{t},$$

when the limit exists in $W$.

## Iterated directional derivatives

**Definition 24** ($n$-th directional derivative)**.** *Let $n \in \mathbb{N}$ and let $v_1, \dots, v_n \in V$. Define iteratively, whenever the expressions exist:*

1. *$D_{v_1} f : \Omega \to W$ by $x \mapsto (D_{v_1} f)_x$;*
2. *for $k \geq 2$, define $D_{v_k} \cdots D_{v_1} f : \Omega \to W$ by*

$$x \mapsto \big(D_{v_k}(D_{v_{k-1}} \cdots D_{v_1} f)\big)_x.$$

*If the resulting map $D_{v_n} \cdots D_{v_1} f$ is defined on a neighbourhood of $a$ and is directionally differentiable at $a$ as required by the iteration above, we define the $n$-th* directional derivative of $f$ at $a$ in directions $v_1, \dots, v_n$ *by*

$$\big(D_{v_n} \cdots D_{v_1} f\big)_a \in W.$$

**Remark 38** (Conceptual meaning)**.** *At each step one forms a new function by taking a first-order approximation along a line. Then one takes a first-order approximation of that new function along another line. Iterating this procedure produces higher-order directional derivatives.*

## Connection to higher Fréchet derivatives

**Proposition 32** ($n$-times Fréchet differentiable implies existence of $n$-th directional derivatives)**.** *Let $n \in \mathbb{N}$ and assume that $f$ is $n$-times Fréchet differentiable at $a$. Then for every $v_1, \dots, v_n \in V$, the iterated directional derivative $\big(D_{v_n} \cdots D_{v_1} f\big)_a$ exists. Moreover,*

$$\big(D_{v_n} \cdots D_{v_1} f\big)_a = (D^n f)_a(v_n, \dots, v_1),$$

*with the order of the arguments as written.*

**Remark 39** (Why the converse can fail)**.** *Even if all iterated directional derivatives $\big(D_{v_n} \cdots D_{v_1} f\big)_a$ exist (for many or all choices of directions), this does not by itself imply that $f$ is $n$-times Fréchet differentiable at $a$. Fréchet differentiability requires a single bounded $n$-linear map $(D^n f)_a : V^n \to W$ together with a uniform $n$-th order approximation valid for all small increments in $V$. Directional derivatives are obtained by restricting to lines and then iterating, so they provide weaker information and need not assemble into such a single multilinear approximation.*

**Remark 40** (Meaning of "uniform" in Fréchet differentiability)**.** *In the definition of Fréchet differentiability at $a$, the approximation is* uniform in the increment $h$*: for every $\varepsilon > 0$ there exists $\delta > 0$ such that the estimate*

$$\|f(a+h) - f(a) - (Df)_a(h)\|_W \leq \varepsilon \, \|h\|_V$$

*holds for* all $h \in V$ *with $0 < \|h\|_V < \delta$ and $a + h \in \Omega$. Thus, once $\varepsilon$ is fixed, the same $\delta$ works simultaneously for every direction of $h$.*

## Repeated directions

**Remark 41** (Repeated direction)**.** *Let $v \in V$ and let $n \in \mathbb{N}$. If one repeats the same direction, then the iterated directional derivative is denoted by*

$$(D_v^n f)_a := (D_v \cdots D_v f)_a.$$

*Let $I \subseteq \mathbb{R}$ be an open interval containing $0$ such that $a + tv \in \Omega$ for all $t \in I$ (for example, choose $I$ as in Proposition 8). Define*

$$\gamma : I \to V, \qquad \gamma(t) = a + tv,$$

*and define the one-variable map*

$$g : I \to W, \qquad g(t) = f(\gamma(t)) = f(a + tv).$$

*If the one-variable derivative $g^{(n)}(0)$ exists in $W$, then*

$$(D_v^n f)_a = g^{(n)}(0).$$

# Higher-Order Partial Derivatives

## Standing conventions

Throughout, $d \in \mathbb{N}$ and $\Omega \subseteq \mathbb{R}^d$ is open (with the Euclidean norm). Let $(W, \|\cdot\|_W)$ be a normed $\mathbb{R}$-vector space and let $f : \Omega \to W$ be a map. Fix $a = (a_1, \ldots, a_d) \in \Omega$. For $i \in \{1, \ldots, d\}$, let $e_i \in \mathbb{R}^d$ denote the $i$-th standard basis vector. All limits in $W$ are taken with respect to $\|\cdot\|_W$.

## Iterated partial derivatives

**Definition 25** (First-order partial derivatives as a family of functions)**.** *Let $i \in \{1, \ldots, d\}$. Assume that $\partial_i f(x)$ exists for every $x$ in some neighbourhood of $a$ (in particular, for $x = a$). Then we obtain a map*

$$\partial_i f : \Omega \to W, \qquad x \mapsto \partial_i f(x),$$

*defined at least on that neighbourhood.*

**Definition 26** (Higher-order (iterated) partial derivative at a point)**.** *Let $n \in \mathbb{N}$ and let $i_1, \ldots, i_n \in \{1, \ldots, d\}$. Assume inductively that the map*

$$\partial_{i_{n-1}} \cdots \partial_{i_1} f$$

*is defined on a neighbourhood of $a$ and that its $i_n$-th partial derivative at $a$ exists. Then we define the $n$-th* order iterated partial derivative of $f$ at $a$ *by*

$$\partial_{i_n} \cdots \partial_{i_1} f(a) := \partial_{i_n}\big(\partial_{i_{n-1}} \cdots \partial_{i_1} f\big)(a) \in W.$$

**Remark 42** (Order convention)**.** *The expression $\partial_{i_n} \cdots \partial_{i_1} f(a)$ is an* iterated *derivative: one first forms the function $\partial_{i_1} f$, then differentiates that function in the $i_2$-direction, and so on, finally taking the $i_n$-th partial derivative at $a$. No commutation of partial derivatives is asserted by this notation.*

## Partial derivatives as directional derivatives

**Proposition 33** (Higher-order partial derivatives as iterated directional derivatives)**.** *Let $n \in \mathbb{N}$ and let $i_1, \dots, i_n \in \{1, \dots, d\}$. Assume the iterated directional derivative*

$$\left(D_{e_{i_n}} \cdots D_{e_{i_1}} f\right)_a$$

*exists (in the sense of the section on higher-order directional derivatives). Then the iterated partial derivative $\partial_{i_n} \cdots \partial_{i_1} f(a)$ exists and*

$$\partial_{i_n} \cdots \partial_{i_1} f(a) = \left(D_{e_{i_n}} \cdots D_{e_{i_1}} f\right)_a.$$

*Conversely, if $\partial_{i_n} \cdots \partial_{i_1} f(a)$ exists (with the neighbourhood hypotheses needed for the iteration), then the iterated directional derivative above exists and the same identity holds.*

**Remark 43** (Interpretation)**.** *Higher-order partial derivatives are obtained by restricting to coordinate lines one step at a time. Equivalently, they are higher-order directional derivatives along the distinguished directions $e_1, \dots, e_d$.*

## Connection to higher Fréchet derivatives

**Proposition 34** ($n$-times Fréchet differentiable implies existence of all $n$-th order partial derivatives)**.** *Let $n \in \mathbb{N}$ and assume that $f$ is $n$-times Fréchet differentiable at $a$. Then for every choice of indices $i_1, \dots, i_n \in \{1, \dots, d\}$, the iterated partial derivative $\partial_{i_n} \cdots \partial_{i_1} f(a)$ exists and satisfies*

$$\partial_{i_n} \cdots \partial_{i_1} f(a) = (D^n f)_a(e_{i_n}, \dots, e_{i_1}),$$

*with the order of the arguments as written.*

**Remark 44** (Coordinate expansion via multilinearity)**.** *Assume $f$ is $n$-times Fréchet differentiable at $a \in \Omega$, so that $(D^n f)_a : (\mathbb{R}^d)^n \to W$ is an $n$-linear map. Let $v_1, \dots, v_n \in \mathbb{R}^d$. For each $r \in \{1, \dots, n\}$ write $v_r$ in the standard basis as*

$$v_r = \sum_{j=1}^{d} (v_r)_j \, e_j,$$

*where $(v_r)_j \in \mathbb{R}$ denotes the $j$-th coordinate of $v_r$. We expand $(D^n f)_a(v_1, \dots, v_n)$ by using linearity in each slot, one slot at a time.*

*First, fixing $v_2, \dots, v_n$ and using linearity in the first variable,*

$$(D^n f)_a(v_1, v_2, \dots, v_n) = \sum_{j_1=1}^{d} (v_1)_{j_1} \, (D^n f)_a(e_{j_1}, v_2, \dots, v_n).$$

*Next, for each term in this sum, fix $e_{j_1}, v_3, \dots, v_n$ and use linearity in the second variable:*

$$(D^n f)_a(e_{j_1}, v_2, v_3, \dots, v_n) = \sum_{j_2=1}^{d} (v_2)_{j_2} \, (D^n f)_a(e_{j_1}, e_{j_2}, v_3, \dots, v_n).$$

*Continuing in the same way through the $n$-th variable, we obtain the full expansion*

$$(D^n f)_a(v_1, \dots, v_n) = \sum_{j_1, \dots, j_n = 1}^{d} (v_1)_{j_1} \cdots (v_n)_{j_n} \, (D^n f)_a(e_{j_1}, \dots, e_{j_n}).$$

*Finally, by the proposition relating higher Fréchet derivatives and iterated partial derivatives (with the fixed order convention),*

$$(D^n f)_a(e_{j_1}, \dots, e_{j_n}) = \partial_{j_1} \cdots \partial_{j_n} f(a),$$

*so the expansion can be written equivalently as*

$$(D^n f)_a(v_1, \dots, v_n) = \sum_{j_1,\dots,j_n=1}^{d} (v_1)_{j_1} \cdots (v_n)_{j_n} \, \partial_{j_1} \cdots \partial_{j_n} f(a).$$

## Symmetry of Mixed Partial Derivatives

**Remark 45** (No automatic symmetry)**.** *Even in finite dimensions, the existence of the iterated partial derivatives does not by itself imply that*

$$\partial_i \partial_j f(a) = \partial_j \partial_i f(a)$$

*or any higher-order commutation identity. Such symmetry requires additional hypotheses.*

**Proposition 35** (When iterated partial derivatives are symmetric)**.** *Let $\Omega \subseteq \mathbb{R}^d$ be open and let $f : \Omega \to W$, where $(W, \|\cdot\|_W)$ is a normed $\mathbb{R}$-vector space. Fix $a \in \Omega$ and an integer $n \geq 2$.*

*Assume that $f$ is $n$-times Fréchet differentiable on a neighbourhood of $a$ and that the $n$-th derivative map*

$$D^n f : \Omega \to B_n(\mathbb{R}^d, W), \qquad x \mapsto (D^n f)_x,$$

*is continuous at $a$ (equivalently: $f$ is of class $C^n$ in a neighbourhood of $a$). Then the $n$-linear map $(D^n f)_a : (\mathbb{R}^d)^n \to W$ is* symmetric*, meaning that for every permutation $\sigma$ of $\{1, \dots, n\}$ and every $v_1, \dots, v_n \in \mathbb{R}^d$ one has*

$$(D^n f)_a(v_1, \dots, v_n) = (D^n f)_a\big(v_{\sigma(1)}, \dots, v_{\sigma(n)}\big).$$

*In particular, for any indices $i_1, \dots, i_n \in \{1, \dots, d\}$ and any permutation $\sigma$,*

$$\partial_{i_n} \cdots \partial_{i_1} f(a) = \partial_{i_{\sigma(n)}} \cdots \partial_{i_{\sigma(1)}} f(a).$$

Warning. *Without a continuity hypothesis on the relevant higher derivatives in a neighbourhood of $a$, iterated partial derivatives need not commute: it can happen that $\partial_i \partial_j f(a) \neq \partial_j \partial_i f(a)$ even when both exist.*

**Remark 46** (Differentiability does not imply continuity of the derivative map)**.** *Let $\Omega \subseteq V$ be open and let $f : \Omega \to W$ be a map between normed spaces. If $f$ is Fréchet differentiable at $a \in \Omega$, then $f$ is continuous at $a$. However, the existence of the derivative at $a$ does* not *imply that the derivative map*

$$Df : \Omega \to B(V, W), \qquad x \mapsto (Df)_x,$$

*is continuous at $a$. Thus the condition $f \in C^1$ is strictly stronger than "$f$ is differentiable": by definition, $f \in C^1$ means that $f$ is Fréchet differentiable at every point of $\Omega$ and that $Df$ is continuous on $\Omega$.*